# Highlights

**Value-Weighted Retention Targeting Concentrates Harm: A Diagnostic and a Harm-Aware Rule for Budget-Constrained Targeting in Online Gambling**

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- Value-weighted targeting can concentrate incentives on harmed players.
- An exact identity links harm exposure to a covariance and the budget.
- Harm weighting costs no operator value with estimated inputs when harm tracks value.
- A hard harm cap needs accurate inputs and binds only on measured harm.
- In the main design, effect-only ranking beats value weighting at tight budgets.

# Value-Weighted Retention Targeting Concentrates Harm: A Diagnostic and a Harm-Aware Rule for Budget-Constrained Targeting in Online Gambling

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**Abstract**

Retention incentives in online gambling are often allocated by ranking players on estimated treatment effect times lifetime value. Under a fixed budget this value-weighted rule maximises operator value, but operator value and player harm can rise together. We show that the harm exposure of any budget-exhausting rule exceeds the population mean by the covariance between treatment and harm divided by the budget, and give a sufficient condition under which value-weighted ranking makes that covariance positive: harm rises with value among players with the same treatment effect. In a simulation with 200 replications of 10,000 players, value-weighted ranking raises mean harm among treated players from 0.421 to 0.476, and the sign of the gap follows the harm-value association. A harm weight on the score brings exposure to 0.414 and, with estimated components and harm rising with value, raises operator value from 16,446 to 19,045. The gain persists (+3,246) when harm depends on the covariates only through structural value, so it needs no direct link between harm and the effect. A hard cap beats the harm weight only with feasible-oracle components, and its ceiling holds only for the measure it is given. In the main design, value weighting wins with feasible-oracle components, but with estimated effects ranking on the effect alone yields more operator value for every learner at tight and moderate budgets. The default T-learner favours value weighting in the null, misspecified-effect and asymmetric scenarios. A feasible oracle using only covariates reaches 73.5% of the hindsight oracle. All parameters are illustrative.

*Keywords:* policy learning, budget-constrained targeting, gambling harm, uplift modelling, responsible gambling

## 1. Introduction

An online gambling operator has a retention incentive, such as a bonus or a free bet, and a budget that covers only a fraction of its active players. The operator must choose whom to reach. Practice ranks players by the product of two estimates: the incentive's effect on retention and the player's lifetime value (Haupt and Lessmann, 2022; Gubela and Lessmann, 2021; Hitsch and Misra, 2024). The rule is attractive because it points the budget at players who both respond and are worth keeping.

In gambling the second factor is not neutral. Revenue is concentrated in a small group of heavy players, and that group carries a large share of gambling problems (Fiedler et al., 2019; Tom et al., 2014). Higher spend is associated with worse financial and health

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outcomes (Muggleton et al., 2021). A rule that ranks on value therefore risks ranking on harm. Regulators treat the risk as real. In Great Britain, licensees must stop marketing and the uptake of new bonus offers once strong indicators of harm appear (Gambling Commission, 2022), and new rules on promotions were announced in March 2025 and take effect in January 2026 (Gambling Commission, 2025). An operator that ranks on value and audits harm afterwards may find that the rule has already pointed its budget at the players the regulator asks it to protect.

This paper studies that problem inside budget-constrained empirical welfare maximisation (Kitagawa and Tetenov, 2018; Athey and Wager, 2021). The operator chooses a rule that treats at most a fraction $\kappa$ of players to maximise operator value, the expected retained value created by the incentive. Operator value is a different quantity from player harm, and the two can move together. With an outcome value attached to each player, the optimal rule ranks by effect times value, which is the standard threshold result applied to a value-weighted outcome rather than a new argument. Our contribution is to ask what that rule does to harm, how to correct it, and what survives when its components are estimated.

*Findings..* The first finding is an identity. For any rule that exhausts the budget, the mean harm of treated players exceeds the population mean by the covariance between treatment and harm divided by the budget. Value-weighted ranking makes that covariance positive whenever harm rises with value among players with the same treatment effect (Proposition 2). In our simulation the estimated rule raises mean harm among treated players from 0.421 to 0.476, a gap of +0.055 (95% CI +0.051 to +0.058). The concentration is conditional: it follows the harm-value association, and the gap changes sign when that association does.

The second finding concerns the remedy. A harm weight on the score brings exposure to 0.414, below the population mean, and with estimated components it raises operator value from 16,446 to 19,045, a difference of +2,598 (95% CI 2,406 to 2,790, Cohen's $d$ 1.89). The gain does not need a direct link between harm and the effect: in a decoupled design, where harm depends on the covariates only through structural value, it persists and grows (Section 6.4). The hard cap is the population optimum for a harm ceiling, but with estimated components it earns less than the harm weight, and it guarantees only the harm that its harm measure sees.

The third finding is that, in the main design, value weighting is an oracle property. With feasible-oracle effects and values it wins at every budget we test. With estimated effects, ranking on the effect alone yields more operator value than value weighting for every learner we test at budgets up to $\kappa = 0.30$. The property is scoped to that design: with the default T-learner, value weighting wins in the null scenario (+5,157), the misspecified-effect scenario (+287) and the asymmetric-moderation scenario (+398) of Appendix Appendix E. The tight-budget penalty of value weighting, and the budget at which it starts to pay, therefore belong to the learner and the design, not to the rule. A feasible oracle that sees only covariates, and not the realised effect and value, is the right benchmark for what estimation can recover, and it reaches 73.5% of the hindsight oracle.

The findings change what an operator should do before deployment. The harm diagnostic is cheap: it needs only a harm measure and the treatment assignment, and it can be computed for any rule, including the rule already in use. Logit noise pulls a harm measure toward 0.5, so the gap measured on it tends to be attenuated relative to the gap on true harm. The harm-aware rule costs no operator value in our simulation with estimated components and harm rising with value, so the usual argument that safety is

paid for in revenue does not hold there. The argument does hold with oracle components, where the harm weight and the hard cap both earn less than the harm-blind rule. We return to these points in Section 7.

*Scope..* Every optimality result is conditional on operator value being multiplicatively separable in treatment effect, value and preference weight (Assumption 2 and Definition 1). The $\hat{\tau} \cdot \hat{V}$ heuristic already takes this form, but it is an assumption. Section 8 gives a plausible operator preference under which it fails. All simulation parameters are illustrative, and no operator data informed them.

*Contributions..* We claim no new optimality result for value weighting. The contributions are a diagnostic, a rule, a benchmark and an attribution.

1. **A harm-concentration diagnostic.** We give the identity that exposure minus the population mean equals the covariance of treatment and harm divided by the budget, and a sufficient condition for a positive gap under value-weighted ranking (Proposition 2). In simulation the estimated rule concentrates harm more than the feasible oracle, and the sign of the gap follows the harm-value association (Section 6.2).
2. **A harm-aware rule and the cost of safety.** We state a soft harm weight (Corollary 1) and a hard harm cap, which is a special case of empirical welfare maximisation with linear constraints (Corollary 2). With estimated components and harm rising with value, the soft weight cuts exposure below the population mean and raises operator value, a decoupled-harm design shows the gain works through structural value, not through a direct harm-effect link, and under a noisy harm measure both rules overshoot their targets (Sections 6.3 and 6.4).
3. **Value weighting as an oracle property of the main design.** With feasible-oracle effects and values, value weighting wins at every budget. With estimated effects, effect-only ranking wins for every learner at budgets up to $\kappa = 0.30$, the crossover budget $\kappa^\star$ exists only for some learners, and learner choice matters: a tuned S-learner, trained on $Y$ and multiplied by $\hat{V}$, outperforms the tuned T-learner (Sections 6.5 and 6.6).
4. **Benchmarking and attribution.** We measure estimation cost against a feasible oracle, which reaches only 73.5% of the hindsight oracle. A Shapley split attributes almost all regret to the effect estimate, plug-in calibration of the preference weight is a tautology, and AIPW calibration costs operator value (Sections 6.5 and 8, Appendix Appendix C).

*Roadmap..* Section 2 reviews related work. Section 3 fixes notation and assumptions. Section 4 states the operator's decision problem, the optimal rule, the regret representation and the cost of a preference weight. Section 5 gives the harm-concentration result and the harm-aware rules. Section 6 reports the simulation study. Section 7 sets out preconditions for responsible deployment. Sections 8 and 9 discuss scope and conclude.

## 2. Related Work

### *2.1. Policy Learning and Constrained Welfare Maximisation*

The optimal treatment rule literature assigns treatment to maximise a population welfare criterion. Manski (2004) derives minimax-regret treatment rules under ambiguity,

and Manski and Tetenov (2007) studies rules for a planner who is averse to risk in aggregate welfare. Kitagawa and Tetenov (2018) formulate empirical welfare maximisation (EWM) over a restricted class of rules, and Athey and Wager (2021) give doubly robust policy learning with regret guarantees of order $n^{-1/2}$ under complexity restrictions on the class. Bhattacharya and Dupas (2012) characterise welfare-maximising assignment under a budget constraint and derive the ranked-threshold structure directly. The rate of Athey and Wager (2021) is conditional on the complexity restriction, so it need not describe flexible learners used in practice. Our optimal rule is the threshold result of these papers applied to a value-weighted outcome (Remark 3).

Several papers add constraints or criteria beyond mean welfare. Sun (2024) studies EWM when the budget cost of a policy must itself be estimated, which covers moment constraints of the kind our hard harm cap imposes. Kitagawa and Tetenov (2021) allow welfare criteria that trade efficiency against equality. Viviano and Bradic (2024) learn targeting rules under fairness constraints on a protected group. Our hard cap (Corollary 2) is a special case of this literature, and we claim no novelty for it. What is new here is the setting: the constrained quantity is the harm of the players an incentive reaches, and the question is how a value-weighted rule behaves against it. The constraint is also unusual in a second way, because it binds only on the harm measure the rule is fed, so measurement error in that measure carries through to the guarantee.

A second strand supplies the estimation and evaluation tools such rules need. Kallus (2018) proposes balanced policy evaluation and learning, Zhao et al. (2012) outcome-weighted learning of treatment rules, and Zhou et al. (2023) offline policy learning with multiple actions. Chernozhukov et al. (2018) give inference for heterogeneous effects with machine learning, Imai and Li (2023) evaluate treatment rules experimentally under a budget on the treated share, and Yadlowsky et al. (2025) evaluate prioritisation rules by rank-weighted average treatment effects. Our evaluation is simulation-based and uses known effects, so it complements rather than uses these estimators.

### *2.2. Value-Weighted and Budgeted Uplift Targeting*

Haupt and Lessmann (2022) study targeting under response-dependent costs and rank players by a value- and cost-weighted treatment effect, of which $\hat{\tau}_i \cdot V_i$ is the cost-homogeneous case. Gubela and Lessmann (2021) propose value-driven evaluation metrics for uplift models, and the profit-uplift literature (Gubela et al., 2020; Fernández-Loría and Provost, 2022; Verbeke et al., 2017) frames targeting as expected-value maximisation. Hitsch and Misra (2024) characterise the optimal policy as a ranking by expected incremental profit. Budgeted ranking by an uplift score also has a long applied lineage in the Qini and profit-uplift literature (Radcliffe, 2007; Gubela et al., 2019; Devriendt et al., 2022), and Devriendt et al. (2022) note that budgeted targeting depends only on the rank order of the score. We claim no novelty for value weighting or for budgeted ranking. The contribution here is to study what value weighting does to harm, and how its performance changes when effect and value are estimated. This prior work evaluates ranking rules on marketing outcomes such as conversion or profit, where a high-value customer is not also a customer the firm has a duty to protect. In gambling that separation fails, which is why the value factor needs scrutiny that the marketing literature has not given it.

### *2.3. Learners for Heterogeneous Effects*

The framework is agnostic to the estimator of the effect $\tau(x)$. We use the T-, S- and X-learners (Künzel et al., 2019), a doubly robust learner (Kennedy, 2023) and the

R-loss (Nie and Wager, 2021) for tuning, and causal forests (Wager and Athey, 2018) are a further option. Devriendt et al. (2018) survey uplift methods and benchmark them on marketing data.

### *2.4. Gambling Harm, Incentives and Regulation*

Gambling revenue is concentrated. Fiedler et al. (2019) find that a small group of problem gamblers accounts for a large share of spending, and Tom et al. (2014) find that a small share of online gamblers accounts for most activity and most problems. Muggleton et al. (2021) link higher gambling spend in bank-transaction data to adverse financial, social and health outcomes. Harm is not confined to problem gamblers: Browne et al. (2016) show that it is spread across risk levels, and Wardle et al. (2019) frame gambling harm as a public-health cost. Harm measurement is contested and imperfect (Delfabbro and King, 2019), which is why we study a noisy harm measure. Behavioural markers in account data can flag risk (Braverman and Shaffer, 2012; Auer and Griffiths, 2023), and Delfabbro et al. (2025) validate such markers against the Problem Gambling Severity Index. We treat a harm score of this kind as an input and do not model how it is produced.

Inducements are a regulated concern. Hing et al. (2017) analyse the structural features of betting inducements for harm minimisation, and Newall et al. (2025) analyse free bets and play-through requirements and propose disclosure of their economic value. In Great Britain the Customer Interaction guidance requires licensees to stop marketing and the uptake of new bonus offers when strong harm indicators are identified (Gambling Commission, 2022). Licence condition SR code 5.1.1 requires incentive terms to be clear and fair and caps wagering requirements at ten times (Gambling Commission, n.d.). Rules announced on 26 March 2025 take effect on 19 January 2026 (Gambling Commission, 2025), within the wider reform set out in the Department for Digital, Culture, Media and Sport (2023) white paper. These rules constrain a value-weighted targeter directly, and they motivate treating harm as part of the targeting problem rather than as a filter applied afterwards.

Outside gambling, retention research shows that targeting by predicted risk is often ineffective and that responsiveness should drive allocation (Ascarza, 2018; Ascarza et al., 2018; Lemmens and Gupta, 2020). Taken together, the literature gives us three inputs: evidence that revenue and harm concentrate on the same players, regulatory rules that bind targeting on harm indicators, and a policy-learning toolkit that can express a harm constraint. We are not aware of work that asks whether value-weighted retention targeting concentrates harm in gambling, or that compares harm-aware rules under estimated components, which is the gap this paper addresses.

## 3. Setup and Notation

Consider $N$ individuals indexed by $i$, each with a pre-treatment covariate vector $X_i \in \mathcal{X} \subseteq \mathbb{R}^d$, a binary treatment $W_i \in \{0,1\}$ (a retention bonus) and a binary outcome $Y_i \in \{0,1\}$ (retention over the evaluation window). The potential outcomes are $Y_i(0)$ and $Y_i(1)$.

**Assumption 1** (Causal identification)**.** *(i)* Unconfoundedness: $(Y_i(0), Y_i(1)) \perp W_i \mid X_i$. *(ii)* Bounded overlap: *there is a constant* $\underline{e} > 0$ *such that* $\underline{e} \leq \mathbb{P}(W_i = 1 \mid X_i = x) \leq 1 - \underline{e}$ *for every* $x \in \mathcal{X}$. *(iii)* SUTVA: *there is no interference between units. (iv)* Consistency: $Y_i = Y_i(W_i)$.

Under Assumption 1 the conditional average treatment effect (CATE) is identified:

$$\tau(x) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x]. \tag{1}$$

In a randomised experiment all four parts hold by design and $\tau(x)$ can be estimated from the observed data directly. In observational data, estimating $\tau(x)$ requires a propensity model or an estimator that absorbs confounding (Section 7).

**Assumption 2** (Incremental operator value)**.** *For each individual $i$, $V_i \geq 0$ is the operator value of retaining $i$ over the evaluation horizon, and $R_i \in [0,1]$ and $H_i \in [0,1]$ are the risk covariate and the harm status. (i)* Pre-treatment: *$V_i$, $R_i$ and $H_i$ are determined before treatment and are not affected by it, so $V_i(1) = V_i(0) = V_i$, and likewise for $R_i$ and $H_i$. (ii)* Retention is the only channel: *treating $i$ changes operator value only through retention, so the incremental operator value of treating $i$ is $\big(Y_i(1) - Y_i(0)\big)V_i$. The (homogeneous) cost of treatment is accounted for by the budget constraint.*

The operator also chooses a measurable preference weight $g : [0,1] \to [0,\infty)$ on the risk covariate, with $g \equiv 1$ the neutral choice. Under Assumption 2, define the *score*

$$s(x) = \mathbb{E}\Big[\big(Y_i(1) - Y_i(0)\big)\, V_i\, g(R_i) \;\Big|\; X_i = x\Big]. \tag{2}$$

If the individual effect $Y_i(1) - Y_i(0)$ is independent of $(V_i, R_i)$ given $X_i$, then $s(x) = \tau(x)\,\mathbb{E}[V_i\, g(R_i) \mid X_i = x]$, and if $V_i$ and $R_i$ are functions of $X_i$ this is the product $\tau(x)\, V(x)\, g(R(x))$. Part (i) is what makes $V_i$ and $R_i$ usable as weights: a value or risk score that the treatment itself moves would introduce post-treatment bias. Part (ii) rules out channels such as a bonus changing how much a retained player later spends; Section 8 discusses response-dependent costs.

**Remark 1** (Pre-treatment requirement)**.** *Lifetime value and churn risk in real systems are usually estimated from behavioural histories that earlier promotions shaped. Using such estimates as $V_i$ or $R_i$ violates Assumption 2(i). Compliance requires cohorts defined before any promotional activity, or auxiliary models restricted to features that pre-date treatment exposure. A feature-provenance audit is therefore a precondition for applying the rule on top of existing value and churn models.*

Players differ in harm status $H_i$. Section 5 defines the conditional mean $\bar{H}(x)$, the operator's pre-treatment harm measure $\hat{H}_i$ and the harm exposure $\eta(\pi)$ of a rule. A treatment rule is a measurable $\pi : \mathcal{X} \to [0,1]$, deterministic when it takes values in $\{0,1\}$, and the budget set is

$$\Pi_\kappa = \Big\{\pi : \mathcal{X} \to [0,1] \;\Big|\; \mathbb{E}[\pi(X_i)] \leq \kappa\Big\}, \qquad \kappa \in (0,1). \tag{3}$$

Estimated quantities carry a hat. Table 1 collects the symbols.

## 4. The Operator's Decision Problem

The operator ranks players by the incremental operator value that treating them adds, subject to the budget. This section defines that objective, gives the optimal rule, represents the cost of estimation error and shows what a preference weight costs.

Table 1: Notation reference.

| Symbol | Meaning | First used |
|---|---|---|
| $X_i \in \mathcal{X}$ | pre-treatment covariate vector | §3 |
| $W_i \in \{0,1\}$ | treatment indicator (distinct from $\mathcal{W}(\pi)$) | §3 |
| $Y_i \in \{0,1\}$ | binary outcome (retention) | §3 |
| $\tau(x)$ | conditional average treatment effect | §3 |
| $V_i \geq 0$ | operator value of retaining $i$ | §3 |
| $R_i \in [0,1]$ | risk covariate (churn probability) | §3 |
| $H_i \in [0,1]$ | player harm status | §3 |
| $g$ | operator preference weight on $R_i$ | §3 |
| $s(x)$ | score $\mathbb{E}[(Y_i(1) - Y_i(0))V_i g(R_i) \mid X_i = x]$ | §3 |
| $\kappa \in (0,1)$ | budget (fraction treatable) | §3 |
| $\Pi_\kappa$ | budget set of treatment rules | §3 |
| $\mathcal{W}(\pi; g)$ | operator welfare of rule $\pi$ | §4 |
| $c^*$, $\pi^*$ | optimal threshold and rule; $\hat{\pi}$ an estimated rule | §4 |
| $\mathcal{R}$ | regret of an estimated rule | §4 |
| $\bar{H}(x)$ | conditional mean harm $\mathbb{E}[H_i \mid X_i = x]$ | §5 |
| $\hat{H}_i$, $\hat{h}(x)$ | operator's harm measure and its conditional mean | §5 |
| $\omega$ | soft harm weight, non-increasing in $\hat{h}$ | §5 |
| $g_{\min}$ | floor of the preference weight $g$ | App. Appendix B |
| $\eta(\pi)$, $\eta_{\max}$ | harm exposure of a rule; ceiling on measured exposure | §5 |
| $\lambda$, $\mu$ | multipliers on the budget and the harm constraint | §5 |

**Definition 1** (Operator welfare)**.** *For a measurable $g : [0,1] \to [0,\infty)$ and a treatment rule $\pi : \mathcal{X} \to [0,1]$ (deterministic rules take values in $\{0,1\}$), the operator welfare of $\pi$ is*

$$\mathcal{W}(\pi; g) = \mathbb{E}\Big[\pi(X_i)\,(Y_i(1) - Y_i(0))\,V_i\,g(R_i)\Big] = \mathbb{E}\Big[\pi(X_i)\,s(X_i)\Big], \tag{4}$$

*where the second equality follows from iterated expectations because $\pi(X_i)$ is a function of $X_i$. We assume $\mathbb{E}|s(X_i)| < \infty$.*

**Proposition 1** (Optimal budget-constrained rule)**.** *Let Assumption 2 hold, let $\kappa \in (0,1)$, and suppose (a) $\mathbb{P}(s(X_i) > 0) \geq \kappa$, and (b) $\mathbb{P}(s(X_i) = c) = 0$ for every $c > 0$. Let $c^* = \inf\{c \geq 0 : \mathbb{P}(s(X_i) > c) \leq \kappa\}$. Then $c^* \geq 0$, $\mathbb{P}(s(X_i) > c^*) = \kappa$, and the rule*

$$\pi^*(x) = \mathbf{1}\{s(x) > c^*\} \tag{5}$$

*maximises $\mathcal{W}(\pi; g)$ over all measurable $\pi : \mathcal{X} \to [0,1]$ with $\mathbb{E}[\pi(X_i)] \leq \kappa$, randomised rules included. Any other maximiser equals $\pi^*$ almost surely. Equivalently, $\pi^*$ treats the top-$\kappa$ fraction of the population ranked by $s$; when $c^* > 0$, ties at the threshold occur with probability zero.*

*Proof sketch.* For any feasible $\pi$,

$$\mathcal{W}(\pi^*; g) - \mathcal{W}(\pi; g) = \mathbb{E}\Big[(\pi^* - \pi)(s - c^*)\Big] + c^*\Big(\kappa - \mathbb{E}[\pi]\Big).$$

The integrand of the first term is non-negative pointwise: $\pi^* = 1 \geq \pi$ where $s > c^*$, and $\pi^* = 0 \leq \pi$ where $s < c^*$. The second term is non-negative because $c^* \geq 0$ and $\mathbb{E}[\pi] \leq \kappa$. Condition (a) gives $c^* \geq 0$ and condition (b) gives $\mathbb{P}(s > c^*) = \kappa$ and almost-sure uniqueness. No constraint qualification is needed; the full argument is in Appendix Appendix A. This is the threshold logic of Kitagawa and Tetenov (2018, Theorem 1) and Bhattacharya and Dupas (2012) applied to the score $s$. □

**Remark 2** (Slack budget)**.** *If $\mathbb{P}(s(X_i) > 0) < \kappa$, the budget does not bind: every rule that treats $\{s > 0\}$, does not treat $\{s < 0\}$, and spends any remaining budget on $\{s = 0\}$ is optimal, so the optimum is not unique.*

**Remark 3** (Reduction to a transformed outcome)**.** *Let $\tilde{Y}_i = Y_i V_i g(R_i)$. By Assumption 2(i), $\tilde{Y}_i(w) = Y_i(w) V_i g(R_i)$, so the CATE of $\tilde{Y}$ is exactly $s(x)$ in (2), and $\mathcal{W}(\pi; g)$ is the unweighted EWM welfare of Kitagawa and Tetenov (2018) for $\tilde{Y}$. Proposition 1 is therefore their threshold result for a transformed outcome, not a new argument. The product form $\tau V g$ is the factorisation of $s$ under the conditional-independence condition stated after Assumption 2; the decomposed rule estimates the factors separately, while a joint learner trained on $\tilde{Y}$ estimates $s$ directly.*

Setting $V_i = 1$ and $g \equiv 1$ recovers the welfare of Kitagawa and Tetenov (2018), and $g \equiv 1$ alone gives the value-weighted welfare behind the practice of Haupt and Lessmann (2022) and Gubela and Lessmann (2021). We call $\mathcal{W}$ *operator welfare* and reserve *player harm* for $H_i$: the results here concern what the operator maximises, and the two can point in different directions.

**Remark 4** (Threshold and ordinal sufficiency)**.** *The threshold $c^*$ is the shadow price of the budget: the minimum score at which a unit of budget is worth spending, and the operator value that one more unit of budget would buy. The rule depends on the scores only through their ranking, so an estimated score that ranks players as the true score does selects the same set. Rank preservation is a strong condition for a product of estimated components, because monotone distortions of the factors do not in general preserve the order of their product.*

**Lemma 1** (Regret representation)**.** *Under the conditions of Proposition 1, let $\hat{\pi} : \mathcal{X} \to \{0, 1\}$ be any deterministic rule with $\mathbb{E}[\hat{\pi}(X_i)] \leq \kappa$, for example a top-$\kappa$ rule on an estimated score fitted on an independent sample. Its regret $\mathcal{R}(\hat{\pi}) = \mathcal{W}(\pi^*; g) - \mathcal{W}(\hat{\pi}; g)$ satisfies*

$$\mathcal{R}(\hat{\pi}) = \mathbb{E}\Big[\, |s(X_i) - c^*| \, \mathbf{1}\{\hat{\pi}(X_i) \neq \pi^*(X_i)\}\Big] + c^*\Big(\kappa - \mathbb{E}[\hat{\pi}(X_i)]\Big), \tag{6}$$

*and the second term vanishes when $\hat{\pi}$ exhausts the budget.*

*Proof.* Apply the decomposition in the proof of Proposition 1 with $\pi = \hat{\pi}$. For binary rules, $(\pi^* - \hat{\pi})(s - c^*)$ equals $s - c^* > 0$ where $\pi^* = 1, \hat{\pi} = 0$, equals $c^* - s \geq 0$ where $\pi^* = 0, \hat{\pi} = 1$, and is zero where the rules agree, which is $|s - c^*| \, \mathbf{1}\{\hat{\pi} \neq \pi^*\}$. □

Regret is driven only by individuals the two rules disagree on, weighted by their distance from the threshold. This is the margin representation behind fast rates for plug-in classifiers (Audibert and Tsybakov, 2007) and the threshold structure of Kitagawa and Tetenov (2018); we use it descriptively and derive no rates.

**Remark 5** (Order-free attribution of regret)**.** *For a decomposed rule ranking by $\hat{\tau}\hat{V}$, let $\mathcal{R}(a, b)$ be the regret of the top-$\kappa$ rule that ranks by $a \cdot b$, measured against the feasible oracle that ranks by $\bar{\tau}\bar{V}$ with $\bar{\tau}(x) = \mathbb{E}[\tau_i \mid X_i = x]$ and $\bar{V}(x) = \mathbb{E}[V_i \mid X_i = x]$, so that $\mathcal{R}(\bar{\tau}, \bar{V}) = 0$. Replacing one component at a time attributes regret in an order-dependent way. The two-player Shapley values average over both orders:*

$$\phi_\tau = \tfrac{1}{2}\Big[\mathcal{R}(\hat{\tau}, \bar{V}) + \mathcal{R}(\hat{\tau}, \hat{V}) - \mathcal{R}(\bar{\tau}, \hat{V})\Big], \qquad \phi_V = \tfrac{1}{2}\Big[\mathcal{R}(\bar{\tau}, \hat{V}) + \mathcal{R}(\hat{\tau}, \hat{V}) - \mathcal{R}(\hat{\tau}, \bar{V})\Big],$$

*and $\phi_\tau + \phi_V = \mathcal{R}(\hat{\tau}, \hat{V})$. A share $\phi_\tau / \mathcal{R}$ above one means that, averaged over orders, swapping in $\hat{V}$ reduces regret, so the value estimate partly offsets error in $\hat{\tau}$.*

**Observation 1** (Preference weights and plug-in calibration)**.** *Let $\pi_g^*$ maximise $\mathcal{W}(\pi; g)$ over the budget set and let $\pi_{\mathbf{1}}^*$ maximise $\mathcal{W}(\pi; \mathbf{1})$. Then $\mathcal{W}(\pi_g^*; \mathbf{1}) \leq \mathcal{W}(\pi_{\mathbf{1}}^*; \mathbf{1})$, because $\pi_g^*$ is feasible for the problem that $\pi_{\mathbf{1}}^*$ solves; under the conditions of Proposition 1 for $g \equiv 1$, equality holds if and only if $\pi_g^* = \pi_{\mathbf{1}}^*$ almost surely. The same argument applies on any sample. Suppose $g$ is chosen from a finite candidate set containing the identity by maximising the plug-in welfare $\widehat{\mathcal{W}}_n(\pi) = n^{-1} \sum_{i=1}^n \pi(X_i)\, \hat{\tau}(X_i) \hat{V}(X_i)$ on a validation sample, where each candidate rule $\pi_g$ treats the $\lfloor \kappa n \rfloor$ units with the largest $\hat{\tau} \hat{V} g(\hat{R})$. The identity candidate treats the $\lfloor \kappa n \rfloor$ units with the largest $\hat{\tau} \hat{V}$, which maximises $\widehat{\mathcal{W}}_n$ over all sets of that size, so the identity attains the maximum on every sample. Plug-in calibration therefore selects the identity (or a candidate with the same plug-in welfare) by construction, and its selection rate carries no evidence about whether $g$ helps. Evidence must come from an objective that is not maximised by construction, such as a doubly robust (AIPW) estimate of $\mathcal{W}(\pi; \mathbf{1})$.*

**Remark 6** (Decomposed and joint scores)**.** *A decomposed rule estimates the factors of $s$ separately, so each model can be audited on its own. A joint learner trained on $\tilde{Y}$ estimates $s$ directly and need not form the product. The S-learner of Section 6 is not of this kind: it is trained on $Y$ and multiplied by $\hat{V}$.*

## 5. Harm Concentration and the Harm-Aware Rule

In gambling, a player's value to the operator is large partly because the player gambles in large volumes, and some of those volumes are harmful. Value weighting can therefore pull incentives toward harmed players although nothing in the score measures harm. Whether it does is an empirical question about the joint distribution of value and harm, and Proposition 2 states exactly which conditions decide it. The exclusion sets, fail-closed defaults and regulatory context that surround any such rule are in Section 7.

Let $H_i \in [0, 1]$ denote true harm, $\bar{H}(x) = \mathbb{E}[H_i \mid X_i = x]$, and for a rule with $\mathbb{E}[\pi(X_i)] > 0$ define the harm exposure $\eta(\pi) = \mathbb{E}[H_i \mid \pi(X_i) = 1]$, the mean harm among treated players.

**Proposition 2** (Harm concentration)**.** *(i) For every deterministic rule $\pi$ with $\mathbb{E}[\pi(X_i)] = \kappa$,*

$$\eta(\pi) - \mathbb{E}[H_i] = \frac{\operatorname{Cov}\Big(\pi(X_i), \bar{H}(X_i)\Big)}{\kappa}. \tag{7}$$

*(ii) Let $\pi(x) = \mathbf{1}\{a(x)\, b(x) > c\}$ with $b \geq 0$, $c \geq 0$ and $\mathbb{E}[\pi(X_i)] = \kappa$, where $(a, b) = (\tau, \bar{V})$ for the oracle rule or $(\hat{\tau}, \hat{V})$ for an estimated rule. Suppose (A) $\mathbb{E}[\bar{H}(X_i) \mid a(X_i), b(X_i)] = \psi(a(X_i), b(X_i))$ for a function $\psi$ that is non-decreasing in its second argument for every value of the first, and (B) $\operatorname{Cov}\Big(\mathbb{P}(\pi(X_i) = 1 \mid a(X_i)),\ \mathbb{E}[\bar{H}(X_i) \mid a(X_i)]\Big) \geq 0$, which holds in particular when $\mathbb{E}[\bar{H}(X_i) \mid a(X_i)]$ does not depend on $a(X_i)$. Then $\eta(\pi) \geq \mathbb{E}[H_i]$. The inequality is strict if, in addition, $\psi(u, \cdot)$ is strictly increasing and $0 < \mathbb{P}(\pi(X_i) = 1 \mid a(X_i) = u) < 1$ for a set of $u > 0$ of positive probability. Reversing the monotonicity in (A) and the sign in (B) reverses the conclusion.*

In words: whenever harm rises with value among players with the same treatment effect, and treatment-effect ranking does not itself steer away from harm, ranking by effect times value concentrates harm among the treated. The condition concerns the ranking components, so it can hold for the estimated rule and fail for the oracle rule, or the reverse. No such association is implied by the targeting problem itself, so concentration

is a property of the data, not of the rule; Section 6.2 illustrates this by sweeping the harm-value alignment $c_{\text{harm}}$ from $-1$ to 1.

*Proof sketch.* (i) is iterated expectations; (ii) applies the law of total covariance conditional on $a(X)$ and Chebyshev's association inequality within each level of $a(X)$; see Appendix Appendix A.

Let $\hat{H}_i \in [0,1]$ be the operator's harm measure, observed before treatment and included in $X_i$, and write $\hat{h}(x) = \mathbb{E}[\hat{H}_i \mid X_i = x]$, which equals $\hat{H}_i$ when $\hat{H}_i$ is a component of $X_i$.

**Corollary 1** (Soft harm weight)**.** *Let $\omega : [0,1] \to [0,1]$ be measurable and non-increasing. Under the conditions of Proposition 1 for $s$, applied to the score $s^\dagger(x) = s(x)\,\omega(\hat{h}(x))$, with $c^\dagger$ its threshold, the rule $\mathbf{1}\{s^\dagger(x) > c^\dagger\}$ maximises $\mathbb{E}[\pi(X_i)\,s^\dagger(X_i)]$ over the budget set. By the argument of Observation 1, its operator welfare $\mathcal{W}(\cdot; g)$ is at most that of the rule without the harm weight.*

The weight is soft: a large enough $s(x)$ clears the threshold whatever $\omega(\hat{h}(x))$ is, so Corollary 1 bounds no exposure. Capping value at a constant $v_{\max}$ is a multiplicative weight of the same form, $\min(1, v_{\max}/V)$, applied to value instead of harm. When a ceiling is required, it must be a constraint. For a ceiling $\eta_{\max}$ on measured exposure, consider

$$\max_{\pi:\mathcal{X}\to[0,1]} \mathbb{E}[\pi(X_i)\,s(X_i)] \quad \text{s.t.} \quad \mathbb{E}[\pi(X_i)] \le \kappa, \qquad \mathbb{E}[\pi(X_i)\,\hat{h}(X_i)] \le \eta_{\max}\,\kappa. \tag{8}$$

**Corollary 2** (Hard harm cap)**.** *Suppose there are multipliers $\lambda^* \ge 0$ and $\mu^* \ge 0$ such that $\pi^\ddagger(x) = \mathbf{1}\{s(x) - \mu^*\hat{h}(x) > \lambda^*\}$ satisfies both constraints of (8), with $\lambda^*\Big(\kappa - \mathbb{E}[\pi^\ddagger]\Big) = 0$ and $\mu^*\Big(\eta_{\max}\kappa - \mathbb{E}[\pi^\ddagger\hat{h}]\Big) = 0$. Then $\pi^\ddagger$ solves (8), and if $\mathbb{P}\Big(s(X_i) - \mu^*\hat{h}(X_i) = \lambda^*\Big) = 0$ every other solution equals $\pi^\ddagger$ almost surely. In particular, $\pi^\ddagger$ attains at least the operator welfare of any rule, including a soft-weight rule, whose measured exposure is at most $\eta_{\max}$ within the same budget. The bound $\mathbb{E}[\hat{h}(X_i) \mid \pi^\ddagger = 1] \le \eta_{\max}$ needs a binding budget, $\mathbb{E}[\pi^\ddagger] = \kappa$, which holds in particular when $\lambda^* > 0$; without it only the slack form $\mathbb{E}[\hat{h}(X_i) \mid \pi^\ddagger = 1] \le \eta_{\max}\,\kappa/\mathbb{E}[\pi^\ddagger]$ follows.*

The proof is the Lagrangian argument of Proposition 1 with a second multiplier; it is in Appendix Appendix A.

We take the existence of $(\lambda^*, \mu^*)$ as a hypothesis; for given multipliers it is a set of moment conditions that can be checked directly. In the simulation the budget is exhausted by top-$\kappa$ selection and $\mu^*$ is the smallest value for which the measured-harm constraint holds, found by one-dimensional search.

*Relation to constrained empirical welfare maximisation..* Problem (8) is empirical welfare maximisation with two linear moment constraints on the rule, applied to the transformed outcome of Remark 3. Bhattacharya and Dupas (2012) derive the threshold structure of welfare-maximising assignment under a budget constraint, and the Lagrangian argument above extends it to a second linear constraint; Sun (2024) studies empirical welfare maximisation when such constraints must themselves be estimated, and Kitagawa and Tetenov (2021) and Viviano and Bradic (2024) add equality and fairness criteria to the targeting objective. Corollary 2 is a special case of this literature, and we claim no novelty for it. What the gambling application adds is that the constrained quantity is a harm measure, which is noisy, so the guarantee needs the qualification in the next remark.

**Remark 7** (The ceiling binds on measured harm)**.** *When the budget binds ($\mathbb{E}[\pi^\ddagger] = \kappa$, in particular when $\lambda^* > 0$), Corollary 2 guarantees $\mathbb{E}[\hat{H}_i \mid \pi^\ddagger = 1] \le \eta_{\max}$, not $\eta(\pi^\ddagger) \le \eta_{\max}$; when it does not bind, the guarantee is the slack form $\eta_{\max}\,\kappa/\mathbb{E}[\pi^\ddagger]$. The difference is the overshoot $\eta(\pi^\ddagger) - \mathbb{E}[\hat{H}_i \mid \pi^\ddagger = 1] = \mathbb{E}[H_i - \hat{H}_i \mid \pi^\ddagger = 1]$. Because the penalty favours players with low $\hat{H}_i$, the treated set over-represents players whose measurement error is negative, so the overshoot tends to be positive when $\hat{H}$ is noisy, even if the error has mean zero in the population. The same selection affects the soft weight, and the dominance statement of Corollary 2 holds for the true score $s$; with estimated scores it need not hold (Section 6.3).*

## 6. Simulation Study

The simulation asks three questions. Does value-weighted ranking concentrate harm, and when? What does a harm-aware rule cost the operator, and what does it guarantee? How do the budget and the learner that estimates the effect shape the value of value weighting? Because we generate the population, we know the true effect, value, churn probability and harm of every player, and every rule is scored on those true quantities. Section 6.1 gives the design, Section 6.2 the harm results, Section 6.3 the harm-aware rules, Section 6.4 a decoupled-harm test, Section 6.5 operator value and the budget diagnostic, and Section 6.6 the choice of learner. Appendices Appendix C to Appendix E hold calibration, the harm sweep and additional scenarios.

### *6.1. Design*

**Instantiation.** The treatment $W_i$ is a binary retention bonus, the outcome $Y_i$ is binary retention over the evaluation window, the operator value $V_i$ is lifetime value, and the risk covariate $R_i$ is the churn probability, written $p_i$ below. The window is an operator choice, and cross-window effects are outside the framework. The design reflects the higher-frequency, variable-stake end of online gaming, with repeated sessions, log-normal weekly spend and a multi-session login cadence. Lottery products have a slower cadence and a less extreme-tailed value distribution. The heavy-tailed value scenario of Appendix Appendix E brackets the tail, and we expect a lower-tail vertical to show a smaller value-weighting penalty.

**Population.** Four pre-treatment covariates are drawn independently: account_age $\sim \mathcal{N}(24, 8)$ clipped to $[1, 60]$, weekly_spend $\sim \exp(\mathcal{N}(3.5, 0.8))$ clipped to $[5, 500]$, login_freq $\sim$ Poisson(4) clipped to $[1, 15]$ and bonus_history $\sim \mathcal{N}(3, 1.5)$ clipped to $[0, 10]$. Let $z.$ denote a feature standardised with moments from a sample of 2,000,000 draws, so that no replication standardises on its own data. Churn probability, value, effect and baseline retention are

$$\begin{aligned}
\operatorname{logit}(p_i) &= -1.0 + 0.04\,(30 - \text{age}_i) - 0.25\,\text{login}_i + 0.01\,(40 - \text{spend}_i) + 0.1\,z_{\text{bonus},i} + \varepsilon_i^{(p)}, \\
&\qquad \varepsilon_i^{(p)} \sim \mathcal{N}(0, 0.4), && (9)\\
V_i &= 10\,\text{spend}_i + 40\,\text{login}_i + 5\,\text{age}_i + \varepsilon_i^{(V)}, \quad \varepsilon_i^{(V)} \sim \mathcal{N}(0, 80), && (10)\\
\tau_i &= \Big(0.06 + 0.10\,z_{\text{login},i} - 0.02\,z_{\text{login},i}^2 - 0.04\,z_{\text{spend},i}\Big)\,4p_i(1 - p_i) + \varepsilon_i^{(\tau)}, \\
&\qquad \varepsilon_i^{(\tau)} \sim \mathcal{N}(0, 0.03), && (11)\\
b_i &= 0.6 + 0.15\,z_{\text{login},i} + 0.05\,z_{\text{spend},i} - 0.08\,z_{\text{age},i} + \varepsilon_i^{(b)}, \quad \varepsilon_i^{(b)} \sim \mathcal{N}(0, 0.05), && (12)
\end{aligned}$$

with $p_i$ clipped to $[0.01, 0.99]$, $V_i$ to $[50, 8000]$, $\tau_i$ to $[-0.20, 0.40]$ and $b_i$ to $[0.05, 0.95]$. Outcomes are $Y_i(0) \sim \text{Bernoulli}(b_i)$ and $Y_i(1) \sim \text{Bernoulli}\Big((b_i + \tau_i)_{[0.01,0.99]}\Big)$. The moderation $4p(1-p)$ is symmetric, which is a simulation convenience: Ascarza (2018) documents faster attenuation at high churn, and Appendix Appendix E reruns the study with an asymmetric moderation. Treatment is a fair coin, $W_i \sim \text{Bernoulli}(0.5)$, so overlap (Assumption 1(ii)) holds with $\underline{e} = 0.5$ and the study does not probe low-propensity regimes. Welfare is scored on the nominal $\tau_i V_i$, which exceeds the clip-adjusted welfare of CATE $\times$ LTV, a nominal-minus-effective difference of +787 (Appendix Appendix E).

**Harm.** True harm is $H_i = \text{sigmoid}\Big(-0.3 + \sigma_L \left(c_{\text{harm}}\, L_i + \sqrt{1 - c_{\text{harm}}^2}\, Z_i\right) + e_i\Big)$, where $L_i = (1.1\, z_{\text{spend},i} + 0.4\, z_{\text{login},i})/\sigma_L$ is the standardised spend-login index, $\sigma_L = \sqrt{1.1^2 + 0.4^2}$, $Z_i \sim \mathcal{N}(0,1)$ and $e_i \sim \mathcal{N}(0, 0.5)$. The alignment $c_{\text{harm}} \in \{-1, -0.5, 0, 0.5, 1\}$ sets whether harm rises with value ($c_{\text{harm}} > 0$), is independent of it (0) or falls with it ($< 0$), and the variance of the harm logit does not depend on $c_{\text{harm}}$. The operator's measure is $\hat{H}_i = \text{sigmoid}\Big(\text{logit}(H_i) + \sigma_h\, \xi_i\Big)$ with $\xi_i \sim \mathcal{N}(0,1)$ and $\sigma_h \in \{0, 0.5, 1\}$. It is a noisy oracle built from true harm, not an estimator: no feature-based model could recover the $Z_i$ and $e_i$ components, so even $\sigma_h = 0$ is more favourable than any real measure.

**Estimation.** Each replication splits the population 70/30 into training and test sets, fits every model on the training set and scores rules on the test set. The default T-learner fits one gradient-boosted classifier per arm (depth 3, 100 trees, learning rate 0.1) and differences the predicted probabilities. The S-learner fits one depth-4 gradient-boosted regressor with treatment as a feature, the X-learner follows Künzel et al. (2019), and the DR-learner regresses the cross-fitted doubly robust pseudo-outcome on $X$ with the known propensity 0.5. Each learner also has a tuned version, chosen by cross-fitted R-loss over a grid of 80 depth, tree-count and learning-rate configurations in 5 pilot replications whose seeds are disjoint from the evaluation seeds (Appendix Appendix E). The default value model is a deliberately wrong proxy: a gradient-boosted regressor trained on $12\,\text{spend}^{0.9} + 35\,\text{login} + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, 200)$, which drops age. The alternative is a well-specified model trained on the true form of Equation (10). The churn model classifies control-arm non-retention, a proxy for $p_i$ rather than $p_i$ itself.

**Rules.** The strategies are Random; CATE only ($\hat{\tau}$); CATE $\times$ LTV ($\hat{\tau}\hat{V}$); Full framework (est.), which multiplies by $g(\hat{p}; \theta_0)$ at the default $\theta_0 = (0.05, 0.40, 0.60, 0.90, 0.20)$ of Appendix Appendix B; Full framework (cal.), with $\theta$ chosen on a validation fold from 22 configurations that include the identity and a near-identity configuration, which is equivalent to the identity because $\hat{p} \geq 0.01$, by an AIPW objective or by the plug-in objective of Observation 1; and the S-learner ranking $\hat{\tau}_S \hat{V}$, which borrows $\hat{V}$. The harm-aware rules use $\hat{H}$. The soft harm weight ranks by $\hat{\tau}\hat{V}\,\omega(\hat{H})$ with $\omega$ falling linearly from 1 to 0 on $[0.6, 0.9]$. The hard harm cap ranks by $\hat{\tau}\hat{V} - \mu\hat{H}$, with $\mu$ the smallest value for which mean $\hat{H}$ among its treated does not exceed a ceiling, and the ceiling equals the soft weight's mean $\hat{H}$ among its treated, so the two rules have equal measured exposure. Each of the three harm-relevant rules (harm-blind, soft weight, hard cap) also runs with feasible-oracle components in place of $\hat{\tau}\hat{V}$.

**Oracles.** The hindsight oracle ranks by the realised $\tau_i V_i$, which no rule can observe. The feasible oracle ranks by $\mathbb{E}[\tau \mid X]\,\mathbb{E}[V \mid X]$, which equals $\mathbb{E}[\tau V \mid X]$ because the effect and value noises are independent, so by Proposition 1 it is the best rule that uses covariates alone. Both use $g \equiv 1$ unless a table says $\theta_0$. Estimation cost is measured against the feasible oracle.

**Replications and inference.** We run 200 replications of 10,000 players with seeds $42 + 7r$. Operator welfare is $\mathcal{W}(\pi; \mathbf{1}) = \sum_{i \text{ treated}} \tau_i V_i$ on the test set, and the reference

budget is $\kappa = 0.30$; Section 6.5 varies $\kappa$ over $\{0.10, 0.20, 0.30, 0.50\}$. We compare rules with paired $t$-tests across replications and report differences with 95% confidence intervals and Cohen's $d$ of the paired differences; $p$-values appear in tables. Holm corrections run within named families and never across them. The families are the eight paired tests of each harm cell, the learner comparisons at each budget, the sixteen effect-only comparisons at each budget (a separate family), the budgets of each effect-source and value-source cell of the crossover analysis (across the whole $\kappa$ grid), the two calibrated-rule comparisons at each budget of Table 5, the three non-degenerate calibrated-versus-identity comparisons of Table C.11, and the comparisons of each robustness scenario. We define $\kappa^\star$ as the smallest $\kappa$ on a 0.025 grid at which $\hat{\tau}\hat{V}$ beats $\hat{\tau}$ alone and keeps beating it at every larger grid budget, a stable up-crossing. Its confidence interval is a percentile bootstrap over replications, censored at the grid ends, and we report how many of the 2,000 draws fall below, inside or above the grid.

### *6.2. Harm concentration*

Table 2 sweeps the alignment $c_{\text{harm}}$ and reports exposure minus the population mean, with 95% confidence intervals. At $c_{\text{harm}} = 1$ the estimated $\hat{\tau}\hat{V}$ rule treats players with mean harm 0.476 against a population mean of 0.421, a gap of +0.055 [+0.051, +0.058]. With feasible-oracle components the gap is +0.033 [+0.032, +0.034], smaller, so estimating the components concentrates harm more than the best covariate-only rule does.

Table 2: Harm exposure of the treated minus the population mean ($\kappa = 0.30$, 95% confidence intervals in brackets). The population mean is that of the row's $c_{\text{harm}}$. The harm-blind gap does not vary with $\sigma_h$ because that rule never reads $\hat{H}$ and exposure is scored on true harm.

| | | Estimated components | | | Feasible-oracle components | | |
|---|---|---|---|---|---|---|---|
| $c_{\text{harm}}$ | $\sigma_h$ | Harm-blind | Soft weight | Hard cap | Harm-blind | Soft weight | Hard cap |
| −1.0 | 0.0 | −0.056 [−0.059, −0.053] | −0.076 [−0.079, −0.073] | −0.076 [−0.079, −0.073] | −0.044 [−0.045, −0.043] | −0.065 [−0.066, −0.064] | −0.065 [−0.066, −0.064] |
| −1.0 | 0.5 | −0.056 [−0.059, −0.053] | −0.079 [−0.082, −0.076] | −0.081 [−0.084, −0.078] | −0.044 [−0.045, −0.043] | −0.069 [−0.070, −0.068] | −0.073 [−0.074, −0.072] |
| −1.0 | 1.0 | −0.056 [−0.059, −0.053] | −0.080 [−0.083, −0.077] | −0.084 [−0.087, −0.081] | −0.044 [−0.045, −0.043] | −0.069 [−0.070, −0.068] | −0.078 [−0.079, −0.077] |
| −0.5 | 0.0 | −0.027 [−0.029, −0.025] | −0.079 [−0.080, −0.077] | −0.079 [−0.080, −0.077] | −0.015 [−0.016, −0.014] | −0.085 [−0.086, −0.085] | −0.086 [−0.087, −0.085] |
| −0.5 | 0.5 | −0.027 [−0.029, −0.025] | −0.079 [−0.081, −0.078] | −0.080 [−0.082, −0.078] | −0.015 [−0.016, −0.014] | −0.085 [−0.086, −0.084] | −0.086 [−0.087, −0.085] |
| −0.5 | 1.0 | −0.027 [−0.029, −0.025] | −0.079 [−0.081, −0.077] | −0.080 [−0.082, −0.078] | −0.015 [−0.016, −0.014] | −0.082 [−0.083, −0.081] | −0.084 [−0.085, −0.083] |
| 0.0 | 0.0 | +0.000 [−0.001, 0.001] | −0.064 [−0.065, −0.062] | −0.064 [−0.065, −0.063] | +0.000 [−0.001, 0.001] | −0.088 [−0.088, −0.087] | −0.088 [−0.089, −0.087] |
| 0.0 | 0.5 | +0.000 [−0.001, 0.001] | −0.065 [−0.066, −0.063] | −0.064 [−0.066, −0.063] | +0.000 [−0.001, 0.001] | −0.087 [−0.088, −0.086] | −0.087 [−0.088, −0.086] |
| 0.0 | 1.0 | +0.000 [−0.001, 0.001] | −0.065 [−0.066, −0.063] | −0.064 [−0.065, −0.063] | +0.000 [−0.001, 0.001] | −0.083 [−0.084, −0.082] | −0.084 [−0.085, −0.083] |
| 0.5 | 0.0 | +0.027 [0.025, 0.028] | −0.043 [−0.045, −0.042] | −0.043 [−0.045, −0.042] | +0.011 [0.011, 0.012] | −0.070 [−0.071, −0.070] | −0.071 [−0.071, −0.070] |
| 0.5 | 0.5 | +0.027 [0.025, 0.028] | −0.045 [−0.046, −0.043] | −0.043 [−0.044, −0.041] | +0.011 [0.011, 0.012] | −0.070 [−0.071, −0.069] | −0.070 [−0.071, −0.070] |
| 0.5 | 1.0 | +0.027 [0.025, 0.028] | −0.046 [−0.047, −0.044] | −0.041 [−0.043, −0.040] | +0.011 [0.011, 0.012] | −0.068 [−0.069, −0.067] | −0.068 [−0.069, −0.068] |
| 1.0 | 0.0 | +0.055 [0.051, 0.058] | −0.008 [−0.010, −0.005] | −0.008 [−0.010, −0.005] | +0.033 [0.032, 0.034] | −0.012 [−0.013, −0.011] | −0.012 [−0.013, −0.011] |
| 1.0 | 0.5 | +0.055 [0.051, 0.058] | −0.011 [−0.013, −0.008] | −0.003 [−0.005, 0.000] | +0.033 [0.032, 0.034] | −0.016 [−0.017, −0.015] | −0.015 [−0.015, −0.014] |
| 1.0 | 1.0 | +0.055 [0.051, 0.058] | −0.015 [−0.017, −0.012] | +0.002 [0.000, 0.005] | +0.033 [0.032, 0.034] | −0.019 [−0.020, −0.019] | −0.018 [−0.019, −0.017] |

The sign of the gap follows $c_{\text{harm}}$. It is negative where harm falls with value, indistinguishable from zero where harm is independent of value, and positive where harm rises with value, consistent with Proposition 2, whose monotonicity condition describes harm rising with value. The sample correlation of harm and value runs from −0.75 at $c_{\text{harm}} = -1$ through +0.00 to +0.80 at $c_{\text{harm}} = 1$. The population mean harm drifts from 0.458 to 0.421 across the sweep, which is why each row is read against its own mean. The mean drifts although the variance of the harm logit is fixed, because $z_{\text{spend}}$ is right-skewed and the mean of a sigmoid of a skewed logit depends on its shape as well as its variance. Concentration is therefore a property of the joint distribution of harm and value, not of the rule alone.

### *6.3. Harm-aware rules and the cost of safety*

With estimated components at $c_{\text{harm}} = 1$ and $\sigma_h = 0$, the soft harm weight cuts exposure to 0.414, below the population mean of 0.421, and raises operator welfare to

19,045 from the harm-blind 16,446 (Figure 1). The paired difference is +2,598 [2,406, 2,790], $d = 1.89$. The hard cap reaches the same measured exposure, 0.413, at welfare 15,701. Hard cap minus soft weight is −3,344 [−3,539, −3,148], $d = -2.38$, so with estimated components the cap is the more expensive rule. Relative to harm-blind welfare, the soft weight has a safety cost of −15.8% (negative means it raises welfare) and the cap one of +4.5%. The soft weight also beats CATE-only ranking, by +2,411 [2,239, 2,583], $d = 1.95$, or +14.5%. The gain is therefore not shrinkage toward effect-only ranking, which Section 6.6 finds pays off for every learner at these budgets. Section 6.4 tests whether it comes from harm carrying information about the effect.

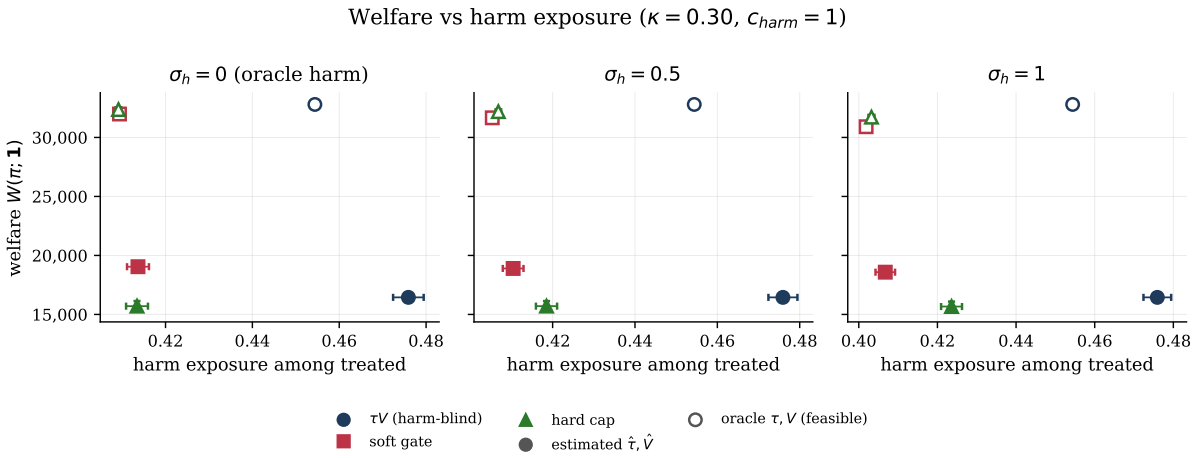



Figure 1: Operator welfare against harm exposure among the treated ($\kappa = 0.30$, $c_{\text{harm}} = 1$), one panel per harm-measure noise $\sigma_h$. Filled markers use estimated components and open circles the feasible oracle.

The order reverses with feasible-oracle components. There the cap earns 32,361 and the soft weight 31,986, against 32,797 for the harm-blind rule; hard cap minus soft weight is +376 [326, 426], $d = 1.05$. The cap's safety cost is +1.3% against +2.5% for the soft weight. The cap is therefore cheaper only with feasible-oracle components, the setting closest to the population statement of Corollary 2. Appendix Appendix D gives welfare for every cell of the sweep.

The ceiling holds only for the harm measure the rule is fed. When $\hat{H}$ is noisy, true exposure exceeds measured exposure for both harm-aware rules. At $\sigma_h = 0.5$ the overshoot is +0.013 [+0.012, +0.014] for the hard cap and +0.005 [+0.004, +0.005] for the soft weight, and at $\sigma_h = 1$ it is +0.043 [+0.041, +0.044] and +0.026 [+0.024, +0.027]. The hard cap exceeds its ceiling on true harm in 100.0% of replications at $\sigma_h = 0.5$ and 100.0% at $\sigma_h = 1$, but the binary rate says less than the size of the overshoot. The mechanism is selection: both harm-aware rules favour players with low $\hat{H}$, which over-represents negative measurement error, and the noise on the logit scale also pulls $\hat{H}$ toward 0.5. With estimated components the soft weight overshoots less than the cap at both noise levels, and it bounds nothing by design (Section 5).

### *6.4. A decoupled-harm test*

The soft weight's gain over harm-blind ranking could be an artefact of the harm design. In the main design harm and the effect both load on spend, so $\hat{H}$ may carry information about $\tau$ beyond $\bar{V}$ that the harm weight exploits. The decoupled design removes that

direct channel. Its harm logit is $-0.3 + \sigma_L\,(c_{\text{harm}}\,\zeta_i + \sqrt{1 - c_{\text{harm}}^2}\,Z_i) + e_i$, where $\zeta_i$ is the standardised noise-free structural value $\bar{V}_i = 10\,\text{spend}_i + 40\,\text{login}_i + 5\,\text{age}_i$, so harm depends on $X_i$ only through $\bar{V}_i$. The population, effect, value, treatment and outcomes are those of the main design, and the logit variance and the draws of $Z_i$, $e_i$ and the measurement noise are unchanged. At $c_{\text{harm}} = 1$ the partial correlation of $\tau$ and the harm logit given $\bar{V}$ is +0.000 in the decoupled design against +0.245 in the main design, and given the realised $V$ it is −0.008 against +0.222. The correlation of harm and realised value is similar in the two designs (+0.82 and +0.80).

Table 3: The soft weight against harm-blind and CATE-only ranking under coupled (main) and decoupled harm ($\kappa = 0.30$, $c_{\text{harm}} = 1$, estimated components, 95% confidence intervals in brackets). Welfare differences are paired over replications. The exposure gap is the soft weight's treated mean true harm minus the population mean harm of that design.

| Harm design | $\sigma_h$ | Soft minus harm-blind | Soft minus CATE only | Exposure gap |
|---|---|---|---|---|
| coupled | 0.0 | +2,598 [2,406, 2,790] | +2,411 [2,239, 2,583] | −0.008 [−0.010, −0.005] |
| decoupled | 0.0 | +3,246 [3,040, 3,452] | +3,059 [2,878, 3,240] | −0.021 [−0.024, −0.019] |
| coupled | 0.5 | +2,450 [2,258, 2,642] | +2,263 [2,092, 2,434] | −0.011 [−0.013, −0.008] |
| decoupled | 0.5 | +3,147 [2,942, 3,351] | +2,959 [2,782, 3,137] | −0.024 [−0.027, −0.022] |
| coupled | 1.0 | +2,135 [1,940, 2,331] | +1,948 [1,774, 2,123] | −0.015 [−0.017, −0.012] |
| decoupled | 1.0 | +2,922 [2,717, 3,127] | +2,735 [2,554, 2,917] | −0.029 [−0.031, −0.026] |

Table 3 shows that the gain persists and grows. At $\sigma_h = 0$ the soft weight beats harm-blind ranking by +2,598 in the main design and by +3,246 [3,040, 3,452] in the decoupled design, $d = 2.20$. Against CATE-only ranking the difference is +2,411 in the main design and +3,059 [2,878, 3,240] in the decoupled design, or +14.5% and +18.4% [+17.0%, +19.7%] of CATE-only welfare, with Holm-adjusted $p < 0.001$ within the cell's eight paired tests. The same holds at $\sigma_h = 0.5$ and 1, and the soft weight's exposure gap is more negative in the decoupled design. With feasible-oracle components the soft weight still beats CATE-only ranking in both designs (+1,138 and +1,482), but it trails harm-blind ranking, by −811 and −468, so decoupling does not remove the cost of safety with accurate components.

The test establishes a limited result. In the decoupled design harm carries no information about $\tau$ beyond $\bar{V}$, so the gain that persists works through $\bar{V}$. A reading consistent with this, which we did not test directly, is that $\hat{H}$ proxies the structural value that the misspecified $\hat{V}$ misses, since the proxy model drops age and flattens spend. The effect is correlated with $\bar{V}$, negatively through the spend term of Equation (11), so part of the information that $\hat{H}$ carries about $\bar{V}$ also bears on $\tau$. The test therefore rules out a direct link between harm and the effect beyond $\bar{V}$ as the source of the gain, and does not rule out every route through which harm informs the ranking.

### *6.5. Operator value, benchmarks and the budget diagnostic*

Table 4 and Figure 2 report operator welfare at $\kappa = 0.30$. The default preference weight costs value: Full framework (est.) earns 13,646 against 16,446 for CATE × LTV, a difference of −2,801 [−2,923, −2,678], or −17.0%. The calibrated rule, Full framework (cal., AIPW), recovers part of that gap (15,566) but stays below the identity. AIPW calibration picks the identity in 13 of 200 replications (6.5%) and costs −881 [−1,217,

$-544]$, $d = -0.37$, against the identity. The plug-in calibrated row equals CATE $\times$ LTV exactly, because the plug-in objective selects the identity in 200 of 200 replications by Observation 1 (Appendix Appendix C).

Table 4: Operator welfare by strategy (10,000 players, 200 replications, $\kappa = 0.30$). Welfare is operator welfare $\mathcal{W}(\pi; \mathbf{1})$ on the test set, SD is the standard deviation across replications, and risk exposure is mean churn probability among the treated. Full framework rows use $\theta_0$ or the calibrated $\theta$ as named, and the S-learner uses the proxy value model.

| Strategy | Welfare | SD | Risk exposure |
|---|---|---|---|
| Random | 5,399 | 1,362 | 0.166 |
| CATE only | 16,634 | 2,962 | 0.153 |
| CATE $\times$ LTV | 16,446 | 3,115 | 0.136 |
| Full framework (est.) | 13,646 | 3,026 | 0.146 |
| Full framework (cal., AIPW) | 15,566 | 4,029 | 0.142 |
| Full framework (cal., plug-in) | 16,446 | 3,115 | 0.136 |
| S-learner | 24,273 | 4,163 | 0.120 |
| Oracle, hindsight ($g$=1) | 44,615 | 807 | 0.138 |
| Oracle, feasible ($g$=1) | 32,797 | 791 | 0.123 |
| Oracle, hindsight ($\theta_0$) | 36,641 | 705 | 0.192 |
| Oracle, feasible ($\theta_0$) | 28,178 | 705 | 0.175 |

The feasible oracle reaches only 73.5% [73.3%, 73.7%] of the hindsight oracle, so about a quarter of hindsight welfare is unreachable from covariates, and we measure estimation cost against the feasible oracle. Against it, the AIPW-calibrated rule, Full framework (cal.), reaches 47.5% [45.8%, 49.2%] in level and 37.0% after subtracting Random and rescaling. CATE $\times$ LTV reaches 50.1% [48.8%, 51.5%] (40.2% normalised) and the S-learner 74.0% [72.3%, 75.8%].

Table 5 varies the budget. CATE $\times$ LTV minus CATE only is $-1{,}482$ $[-1{,}632, -1{,}332]$ at $\kappa = 0.10$, $-842$ $[-998, -687]$ at 0.20, $-187$ $[-319, -55]$ at 0.30, and $+195$ [133, 256] at 0.50. Value weighting hurts at tight budgets and helps at generous ones. With feasible-oracle effects and values it helps at every budget: at $\kappa = 0.10$ the difference is $+1{,}441$ [1,383, 1,499], $d = 3.46$.

Table 5: Operator welfare by budget. CATE only and CATE $\times$ LTV use the default T-learner with the proxy value model; Cal. is Full framework (cal., AIPW); the oracle is the feasible oracle with $g \equiv 1$. Holm $p$-values test Cal. against the named strategy.

| $\kappa$ | CATE only | CATE $\times$ LTV | Cal. (AIPW) | Oracle (feasible) | Holm $p$ vs CATE only | Holm $p$ vs CATE $\times$ LTV |
|---|---|---|---|---|---|---|
| 0.100 | 4,766 | 3,284 | 3,155 | 13,233 | <0.001 | 0.143 |
| 0.200 | 10,750 | 9,908 | 9,363 | 23,864 | <0.001 | <0.001 |
| 0.300 | 16,634 | 16,446 | 15,566 | 32,797 | <0.001 | <0.001 |
| 0.500 | 26,740 | 26,935 | 24,987 | 46,046 | <0.001 | <0.001 |

The crossover budget $\kappa^\star$ is where this penalty ends (Table 6), and it belongs to the learner. Crossing every effect source with every value source isolates its origin. With

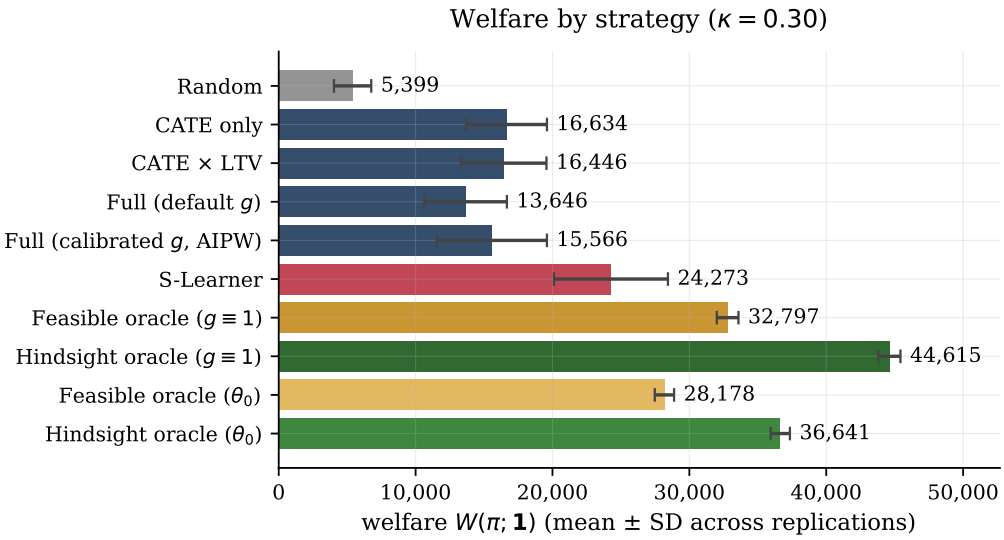



Figure 2: Operator welfare by strategy ($\kappa = 0.30$). Bars show mean welfare and whiskers the standard deviation across replications; the oracles appear as feasible and hindsight pairs.

feasible-oracle effects and values there is no crossing: value weighting leads at every grid budget, and $\kappa^\star$ is $< 0.050$ with a crossing in 0.0% of draws. With the T-learner and the proxy value model $\kappa^\star$ is 0.350 [0.325, 0.375], with a crossing in 100.0% of draws. With the X-learner and the proxy it is 0.425 [0.325, $> 0.500$], crossing in 66.9% of draws, and with the S-learner it never crosses within the grid ($> 0.500$). For the T-learner with feasible-oracle or well-specified values the point estimate sits above the grid and only 43.2% and 24.9% of draws cross inside it. Our explanation is a hypothesis we did not test: multiplying a noisy effect estimate by a value estimate with large spread lets the value estimate dominate the ranking in the tail where the effect is least well measured, and a smoother value estimate or a less noisy effect estimate weakens that.

Table 6: Crossover budget $\kappa^\star$ by effect source and value source. $\kappa^\star$ is the smallest budget on a 0.025 grid from which $\hat{\tau}\hat{V}$ beats $\hat{\tau}$ alone at every larger grid budget; $<$ and $>$ mark values censored at the grid ends. Below, Crossing and Above count the 2,000 bootstrap draws whose $\kappa^\star$ lies below the grid (value weighting leads throughout), inside it, or above it (effect-only ranking leads throughout). Share is the proportion of draws that cross, and Sign changes counts sign changes of the point difference curve.

| $\tau$ source | $V$ source | Below | Crossing | Above | $\kappa^\star$ | 95% CI | Share | Sign changes |
|---|---|---|---|---|---|---|---|---|
| oracle (feasible) | oracle (feasible) V | 2,000 | 0 | 0 | <0.050 | [<0.050, <0.050] | 0.0% | 0 |
| oracle (feasible) | misspecified V | 2,000 | 0 | 0 | <0.050 | [<0.050, <0.050] | 0.0% | 0 |
| oracle (feasible) | well-specified V | 2,000 | 0 | 0 | <0.050 | [<0.050, <0.050] | 0.0% | 0 |
| T | oracle (feasible) V | 0 | 865 | 1,135 | >0.500 | [0.450, >0.500] | 43.2% | 0 |
| T | misspecified V | 0 | 2,000 | 0 | 0.350 | [0.325, 0.375] | 100.0% | 1 |
| T | well-specified V | 0 | 497 | 1,503 | >0.500 | [0.475, >0.500] | 24.9% | 0 |
| X | oracle (feasible) V | 0 | 0 | 2,000 | >0.500 | [>0.500, >0.500] | 0.0% | 0 |
| X | misspecified V | 0 | 1,338 | 662 | 0.425 | [0.325, >0.500] | 66.9% | 1 |
| X | well-specified V | 0 | 0 | 2,000 | >0.500 | [>0.500, >0.500] | 0.0% | 0 |
| S | oracle (feasible) V | 0 | 0 | 2,000 | >0.500 | [>0.500, >0.500] | 0.0% | 0 |
| S | misspecified V | 0 | 0 | 2,000 | >0.500 | [>0.500, >0.500] | 0.0% | 0 |
| S | well-specified V | 0 | 0 | 2,000 | >0.500 | [>0.500, >0.500] | 0.0% | 0 |

*6.6. Learner choice*

Table 7 compares the T-, X-, S- and DR-learners, default and tuned, under both value models and with no value model. Effect-only ranking beats $\hat{\tau}\hat{V}$ in welfare for every learner, tuning and value model at every budget up to $\kappa = 0.30$. The difference is Holm-significant in all but one of these cells, the default DR-learner with the proxy value model at 0.30, where it is +102 [−33, 237]. For the default T-learner with the proxy value model the gain from dropping $\hat{V}$ is +1,482, +842 and +187 at $\kappa = 0.10$, 0.20 and 0.30, and for the tuned S-learner it is +3,372 [2,815, 3,928] at 0.30, $d = 0.84$. These tests form their own Holm family of sixteen comparisons per budget, separate from the learner comparisons in the table. At $\kappa = 0.50$ the default T-, X- and DR-learners with the proxy value model reverse sign (−195 [−256, −133], −20 and −79), and only the T-learner's reversal is Holm-significant. Value weighting is therefore an oracle property of the main design: it wins with feasible-oracle components, and with estimated effects it needs a generous budget to pay for some learners and never pays for the S-learner on the grid. The property does not extend to every scenario, because with the default T-learner value weighting wins in the null (+5,157), misspecified-effect (+287) and asymmetric-moderation (+398) scenarios of Table E.13.

Table 7: Learner comparison at $\kappa = 0.10$ and 0.30 (200 replications). Value model "none" ranks on the estimated effect alone. Holm $p$ tests each cell against T default with the same value model. Shapley is the order-free share of regret against the feasible oracle attributed to the effect estimate, with a 95% confidence interval. The DR-learner equals the R-learner under the known propensity 0.5.

| Learner | Value model | $\kappa = 0.10$ | Holm $p$ | $\kappa = 0.30$ | Holm $p$ | Shapley CATE share |
|---|---|---|---|---|---|---|
| T tuned | misspecified V | 3,125 | 1.000 | 16,941 | 0.233 | 100.6% [100.2%, 101.1%] |
| T tuned | well-specified V | 2,950 | 1.000 | 16,605 | 0.216 | 100.1% [100.0%, 100.2%] |
| T tuned | none (CATE only) | 6,211 | <0.001 | 18,760 | <0.001 | – |
| T default | misspecified V | 3,284 | – | 16,446 | – | 101.1% [100.9%, 101.3%] |
| T default | well-specified V | 3,033 | – | 15,914 | – | 99.9% [99.8%, 100.0%] |
| T default | none (CATE only) | 4,766 | – | 16,634 | – | – |
| X tuned | misspecified V | 6,869 | <0.001 | 22,741 | <0.001 | 101.8% [101.3%, 102.3%] |
| X tuned | well-specified V | 6,541 | <0.001 | 22,243 | <0.001 | 100.1% [99.9%, 100.3%] |
| X tuned | none (CATE only) | 8,854 | <0.001 | 24,218 | <0.001 | – |
| X default | misspecified V | 5,386 | <0.001 | 20,212 | <0.001 | 101.2% [101.0%, 101.5%] |
| X default | well-specified V | 5,035 | <0.001 | 19,729 | <0.001 | 99.9% [99.8%, 100.0%] |
| X default | none (CATE only) | 6,612 | <0.001 | 20,456 | <0.001 | – |
| S tuned | misspecified V | 5,949 | <0.001 | 22,437 | <0.001 | 102.0% [101.5%, 102.5%] |
| S tuned | well-specified V | 5,606 | <0.001 | 21,929 | <0.001 | 99.9% [99.7%, 100.1%] |
| S tuned | none (CATE only) | 8,768 | <0.001 | 25,809 | <0.001 | – |
| S default | misspecified V | 7,585 | <0.001 | 24,273 | <0.001 | 100.6% [100.2%, 101.1%] |
| S default | well-specified V | 7,294 | <0.001 | 24,015 | <0.001 | 99.9% [99.8%, 100.1%] |
| S default | none (CATE only) | 8,874 | <0.001 | 25,351 | <0.001 | – |
| DR tuned | misspecified V | 3,004 | 0.954 | 17,639 | 0.0203 | 101.9% [101.4%, 102.4%] |
| DR tuned | well-specified V | 2,712 | 0.887 | 16,956 | 0.0668 | 100.1% [100.0%, 100.2%] |
| DR tuned | none (CATE only) | 7,352 | <0.001 | 21,693 | <0.001 | – |
| DR default | misspecified V | 3,791 | <0.001 | 16,758 | 0.0879 | 101.1% [100.9%, 101.2%] |
| DR default | well-specified V | 3,454 | <0.001 | 16,239 | 0.0668 | 99.9% [99.8%, 100.0%] |
| DR default | none (CATE only) | 4,715 | 1.000 | 16,861 | 0.216 | – |

Under R-loss tuning the T-learner chooses a heavily shrunk depth-1 fit; its estimated

heterogeneity is strongly attenuated. The tuned T- and DR-learner fits sit at the grid edge toward more shrinkage, and the S-learner at the edge toward less (Appendix Appendix E); the evidence is consistent with a weak-signal regime rather than under-tuning, but a wider grid could change the magnitudes. With tuned learners and the proxy value model, the S-learner earns +5,496 [4,651, 6,341] more than the T-learner at $\kappa = 0.30$, $d = 0.91$, or +32.4% [+26.2%, +38.6%]. The DR-learner equals the R-learner here because with a known constant propensity 0.5 the pseudo-outcomes coincide and the R weights are constant. Three caveats apply to tuning. The spread of the chosen R-loss across learners is only 0.00005, tuning used only 5 pilot replications, and R-loss rewards calibration of the effect estimate, not the ranking that targeting needs. The highest welfare of any estimated rule is the S-learner ranked on the effect alone (25,809 tuned, 25,351 default), an S-learner, one model trained on $Y$ with treatment as a feature, used without a value model; we have not tested it against the other learner and rule cells, so this is descriptive.

The value model matters in a way that contradicts a natural expectation. The proxy value model beats the well-specified one: CATE × LTV with the well-specified model earns −532 [−573, −491] relative to the proxy, $d = -1.81$. A hypothesis consistent with the variance-amplification argument above is that the proxy, which drops age and flattens spend, gives a smoother value estimate and so amplifies less of the effect estimate's noise. The Shapley split attributes essentially all regret to the effect estimate, 99.9% with the well-specified value model and 101.1% with the proxy. A share above 100% means that, averaged over orders, swapping in $\hat{V}$ slightly offsets error in $\hat{\tau}$.

## 7. Responsible Deployment

Sections 5 and 6 describe what the rules do to harm in a simulation. This section sets out what an operator needs in place before any of them is used on a live campaign. The measures below are preconditions, because a value-weighted score does not detect harm on its own.

*The harm proxy..* The harm measure $\hat{H}$ in this paper is a score built from behavioural markers in account data, such as deposit velocity, escalating session length and loss chasing (Braverman and Shaffer, 2012; Auer and Griffiths, 2023; Delfabbro et al., 2025). We use behavioural markers because harm is not confined to players who screen positive on a clinical instrument. It is spread across risk levels (Browne et al., 2016), and its measurement is contested (Delfabbro and King, 2019). The Problem Gambling Severity Index is a survey instrument that few players in an operator's base have completed, so it can validate a marker score (Delfabbro et al., 2025) but cannot serve as the operator's exclusion list. The proxy is a population-level input to targeting, to be audited against harm outcomes the operator can confirm, and not a diagnosis of any individual.

*Order of operations..* Players who have self-excluded, or whom the operator's customer-interaction process has flagged for harm indicators, form an exclusion set $\mathcal{E}$ that is removed before any score is computed. The rule then ranks the eligible players only, $\mathcal{S}^* = \arg\,\text{top-k}_{i \notin \mathcal{E}}\, s_i$. The harm weight of Corollary 1 sits inside the objective, so that harm trades against value for every player who remains. Filtering after ranking is no substitute, because the ranking has already spent the budget. Two defaults follow. The system fails closed: when harm signals are missing for part of the population, the high-value tail of that part is excluded and not scored harm-blind. Exclusion and harm signals are re-derived at every campaign, because a player outside $\mathcal{E}$ at one round may have

crossed a threshold by the next. For the preference weight of Appendix Appendix B we recommend the floor $g_{\min} = 0$ in live use, so that the highest-churn players receive no extra weight. The simulation uses $g_{\min} = 0.20$ for contrast, and any positive floor needs explicit justification, because the highest-churn players are the most likely to be disengaging for reasons connected with harm.

*Soft weight or hard cap..* With estimated components the recommended harm-aware rule is the soft harm weight. It lowers exposure and, in our simulation with estimated components and harm rising with value, does not cost operator value (Section 6.3). A hard cap earns more only when its components are accurate, and its ceiling holds only for the harm measure it is fed (Remark 7). Where a guarantee is required, use the exclusion set, which is the limit of the cap at a ceiling of zero, or a cap on a harm measure that has been audited.

*Harm-model error..* The harm measure is itself an estimate, and Section 6.3 shows what noise in it does. At $\sigma_h = 0.5$, true exposure exceeds measured exposure by +0.013 [+0.012, +0.014] under the hard cap and by +0.005 [+0.004, +0.005] under the soft weight. At $\sigma_h = 1$ the overshoots are +0.043 [+0.041, +0.044] and +0.026 [+0.024, +0.027], and the cap exceeds its ceiling on true harm in 100.0% of replications. An operator should therefore set the operational ceiling below the target by the expected overshoot, and estimate that overshoot on a holdout by comparing the marker score with confirmed harm outcomes. These sizes belong to our simulated noise model, and an operator's own overshoot has to be measured.

*Evaluation and monitoring..* Operator value and harm should be re-estimated each campaign with a doubly robust (AIPW) estimator (Dudik et al., 2011) on a randomised holdout, which keeps the propensities known; counterfactual risk minimisation (Swaminathan and Joachims, 2015) is an alternative. The treatment-effect estimates used in the score can serve as the outcome model, so targeting and evaluation share components. Players in $\mathcal{E}$ are excluded from the holdout as well as from the ranking, because randomising an at-risk player into treatment is what the safeguards forbid. The holdout therefore supports estimates only for the eligible population, and its membership is re-derived each round together with $\mathcal{E}$.

*Regulatory mapping..* Table 8 sets the framework against the requirements most relevant in Great Britain. The Gambling Commission announced its new rules on promotions on 26 March 2025, and they take effect on 19 January 2026 (Gambling Commission, 2025). Inducements are also a recognised harm-minimisation concern in the research literature (Hing et al., 2017; Newall et al., 2025). The table records what the framework addresses and is not a compliance assessment. The methods can also be misused to direct incentives at harmed players, which is why the dual-use statement below ties publication to these safeguards.

## 8. Discussion

### *8.1. Why Estimated Value Weighting Does Not Pay Here*

The oracle result and the estimated result point in opposite directions, and the difference is estimation. Lemma 1 locates regret in players misranked near the threshold; heuristically, in a product score the error in $\hat{\tau}$ is scaled by $\hat{V}$, and $V$ varies by orders of magnitude, so noise in $\hat{\tau}$ is amplified before it ranks players. Effect-only ranking avoids

Table 8: Regulatory requirements in Great Britain and how the framework addresses them.

| Requirement | Source | How the framework addresses it |
|---|---|---|
| Stop marketing and the uptake of new bonus offers once strong indicators of harm are identified | Customer interaction guidance for remote licensees, Requirement 10 (Gambling Commission, 2022); in effect from 31 October 2023 | Addressed in part. The exclusion set and the fail-closed default stop incentives to flagged players before ranking. Flagging depends on the operator's indicators, which the framework does not supply. |
| Wagering requirements capped at 10 times; one product type per incentive; terms clear, transparent and fair | SR code 5.1.1, Rewards and bonuses (Gambling Commission, n.d.) | Not addressed. The framework decides who receives an incentive and not its terms. Compliant terms are a precondition outside the model, and operator value in the simulation ignores wagering terms. |
| High-value or VIP incentives consistent with the licensing objectives | SR code 5.1.1, Rewards and bonuses (Gambling Commission, n.d.) | Addressed in part. Value weighting is the practice this requirement bears on. The diagnostic of Proposition 2 tests whether a rule concentrates incentives on harmed players, and the harm weight reduces the concentration. Neither establishes consistency with the licensing objectives. |
| Revised rules on promotions, announced 26 March 2025 and in force 19 January 2026 | Gambling Commission news article (Gambling Commission, 2025) | Not mapped. We take the dates from the announcement and not its provisions, and an operator must check its offer design against the final rule text. |
| Harm-focused reform of online gambling, including affordability checks | High Stakes white paper, CP 835, April 2023 (Department for Digital, Culture, Media and Sport, 2023) | Not addressed. The framework uses no affordability data. Affordability signals can enter the harm measure where an operator holds them. |

that amplification, and the Shapley split puts nearly all regret on $\hat{\tau}$ (Section 6.6). The value model contributes in an unexpected direction: the well-specified value model earns $-532$ [$-573$, $-491$] relative to the proxy, $d = -1.81$. Our reading, which we did not test, is that the proxy gives a smoother value estimate and so amplifies less of the effect estimate's noise. The tight-budget penalty of value weighting, and the budget at which it starts to pay, therefore belong to the learner and the design, not to the rule.

### *8.2. Calibrating the Preference Weight*

No preference weight we calibrate beats the identity. Observation 1 covers oracle rules only: it shows that no $g$ raises the welfare of an oracle-optimal rule, and that plug-in calibration selects the identity by construction, in 200 of 200 replications. It does not cover estimated rules, and Remark 10 concedes that a $g$ that mirrors a structural interaction the effect model misses can help an estimated rule. That rate is a consistency check on the theory and not a finding. AIPW calibration selects the identity in 13 replications, and the AIPW-calibrated rule earns $-881$ [$-1{,}217$, $-544$] relative to the identity. The operator sets $g$ from preferences, and calibration prices the choice (Appendices Appendix B and Appendix C).

### *8.3. Scope*

Every optimality result is conditional on operator value being multiplicatively separable in effect, value and preference weight (Definition 1). That is the form the $\hat{\tau}\hat{V}$ heuristic already takes, and it is what permits a decomposition into separately auditable components. It fails if, for example, an operator's willingness to take risk exposure varies with customer value, so that the preference factor is $g(R, V)$ and the score no longer factors; such an operator needs a more general objective. The binary outcome restricts the framework to the extensive margin. A continuous outcome changes the notation and not the results, and $V$ may then be redundant. The auxiliary components must be pre-treatment (Remark 1), which sequential deployment weakens because later-period value and churn reflect earlier treatments. Costs are homogeneous here, but the realised cost of a bonus depends on the player's response, and Haupt and Lessmann (2022) derive a profit rule for that case, which we do not simulate. SUTVA rules out spillovers between players who interact.

Harm is treated as pre-treatment, so the incentive's own effect on harm is not modelled. A retention incentive that raises play among harmed players would add to the concentration shown here. The evidence is a simulation with illustrative parameters and no operator data. The public uplift benchmarks cannot test the central claims, because they carry neither a per-player value nor a harm measure, and genuine validation needs operator data with both.

### *8.4. Extensions*

Natural extensions are multi-armed and sequential treatment with bonus sizing, finite-sample regret bounds for the three-component product score in the manner of Athey and Wager (2021), and fairness criteria within the budget-constrained problem (Kitagawa and Tetenov, 2021; Viviano and Bradic, 2024).

## 9. Conclusion

Ranking retention incentives by estimated effect times value is not neutral about harm. For any rule that exhausts the budget, the harm exposure of the treated exceeds the

population mean by the covariance between treatment and harm divided by the budget, and value-weighted ranking makes that covariance positive when harm rises with value among players with the same treatment effect. In our simulation the gap for the estimated rule is +0.055, and it changes sign with the harm-value association. A harm weight on the score removes the concentration without costing operator value when harm rises with value and its components are estimated, and a decoupled-harm test shows the gain does not come from a direct link between harm and the effect beyond structural value. A hard cap is the population optimum for a ceiling, but it needs accurate inputs and guarantees only the harm that its measure sees, so with estimated components we recommend the soft weight and, where a guarantee is required, an audited harm measure and an exclusion set (Section 7).

In the main design, value weighting is an oracle property. With feasible-oracle effects and values it wins at every budget we test, and with estimated effects ranking on the effect alone yields more operator value for every learner at budgets up to $\kappa = 0.30$. The property is scoped to that design, because with the default T-learner value weighting wins in the null (+5,157), misspecified-effect (+287) and asymmetric-moderation (+398) scenarios. A feasible oracle that sees only covariates reaches 73.5% of the hindsight oracle, and it is the benchmark against which estimation cost should be read. All results come from a simulation with illustrative parameters and no operator data. Validating the diagnostic and the harm-aware rule needs operator data with observed lifetime value and a harm measure, and any deployment needs the safeguards of Section 7 whatever that validation shows.

## CRediT authorship contribution statement

**Mark Stent:** Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Writing - original draft, Writing - review and editing.


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This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.


## Declaration of competing interest

The author is employed in the gambling and lottery industry. The framework studied here is not deployed at the author's employer, and there are no plans to deploy it there. The employer did not review this paper. No operator data informed any parameter of the data-generating process; all parameters are illustrative. The views expressed are the author's own.

## Dual-use statement

The methods in this paper can be used to direct incentives more efficiently at players who are valuable to an operator, and Section 5 shows that such targeting can concentrate on harmed players. We publish the diagnostic and the harm-aware rule so that operators, regulators and auditors can detect and limit this effect. We do not support deploying value-weighted targeting without the safeguards of Section 7.

## Ethics statement

This study involves no human participants, no personal data and no experiments on individuals. All results come from a synthetic, fully specified simulation (Section 6), and no real player or customer data was used, so no ethics-board approval was required. Because promotional targeting in online gambling can harm players, the paper reports that value-weighted targeting can concentrate incentives on harmed players (Section 6.2) and treats an exclusion set, a harm-aware score and a fail-closed default as preconditions for any deployment (Section 7).

## Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude (Anthropic) to review the manuscript, refactor and test the simulation code, check references, and edit text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

## Data and code availability

This study uses no empirical data, and every reported quantity is generated by simulation. The code is in the public repository `https://github.com/markstent/bonus-allocation-framework`. It holds the five simulation studies (main comparison, harm sweep, budget analysis, learner comparison and robustness scenarios), the table and macro generator that writes every number in this paper into generated files, the figure generator, and a run manifest that records the git commit, package versions and file hashes. The repository tracks the final run's manifest and result JSONs in `results/`, and `python src/run_all.py` reproduces them. Running `src/make_tex.py` on the results regenerates every number, table and figure. The code is released under the MIT licence and the paper under CC BY 4.0. Each replication $r$ uses the fixed seed $42 + 7r$, so the study is deterministic under the pinned dependencies.

## Appendix A. Proofs

*Proof of Proposition 1*

*Proof of Proposition 1.* Write $F(c) = \mathbb{P}(s(X_i) > c)$, which is non-increasing and right-continuous, with $F(c) \to 0$ as $c \to \infty$. By (b), $F$ is continuous on $(0, \infty)$. *Threshold.* If $F(0) = \kappa$, then $c^* = 0$ and $F(c^*) = \kappa$. If $F(0) > \kappa$, right-continuity gives $F(c) > \kappa$ for small $c > 0$, so $c^* > 0$; continuity of $F$ at $c^*$, together with $F(c) > \kappa$ for $c < c^*$ and $F(c) \leq \kappa$ for $c > c^*$, gives $F(c^*) = \kappa$. Condition (a) excludes $F(0) < \kappa$, so in both cases $c^* \geq 0$ and $\mathbb{E}[\pi^*(X_i)] = \kappa$. *Optimality.* For any measurable $\pi : \mathcal{X} \to [0, 1]$ with $\mathbb{E}[\pi(X_i)] \leq \kappa$, adding and subtracting $c^* \, \mathbb{E}[\pi^* - \pi]$ gives

$$\mathcal{W}(\pi^*; g) - \mathcal{W}(\pi; g) = \mathbb{E}\Big[(\pi^*(X_i) - \pi(X_i))(s(X_i) - c^*)\Big] + c^*\Big(\kappa - \mathbb{E}[\pi(X_i)]\Big).$$

On $\{s > c^*\}$, $\pi^* - \pi = 1 - \pi \geq 0$; on $\{s < c^*\}$, $\pi^* - \pi = -\pi \leq 0$; on $\{s = c^*\}$ the factor $s - c^*$ is zero. The integrand is therefore non-negative, and the second term is non-negative because $c^* \geq 0$. *Uniqueness.* If $\pi$ is also optimal, both terms vanish. The first gives $\pi = \pi^*$ almost surely on $\{s \neq c^*\}$. If $c^* > 0$, (b) gives $\mathbb{P}(s = c^*) = 0$ and we are done. If $c^* = 0$, then $\pi = 1$ almost surely on $\{s > 0\}$, so $\mathbb{E}[\pi] \geq F(0) = \kappa$; feasibility then forces $\pi = 0$ almost surely on $\{s = 0\}$, so again $\pi = \pi^*$ almost surely. □

*Proof of Proposition 2*

*Proof.* (i) Because $\pi(X_i)$ is a function of $X_i$, iterated expectations give $\mathbb{E}[\pi H] = \mathbb{E}[\pi \bar{H}]$ and $\mathbb{E}[H] = \mathbb{E}[\bar{H}]$. Hence $\eta(\pi) = \mathbb{E}[\pi \bar{H}]/\kappa$, and subtracting $\mathbb{E}[H] = \mathbb{E}[\pi]\,\mathbb{E}[\bar{H}]/\kappa$ gives (7). (ii) Write $A = a(X_i)$ and $B = b(X_i)$. By the law of total covariance,

$$\mathrm{Cov}(\pi, \bar{H}) = \mathbb{E}\Big[\mathrm{Cov}(\pi, \bar{H} \mid A)\Big] + \mathrm{Cov}\Big(\mathbb{E}[\pi \mid A],\ \mathbb{E}[\bar{H} \mid A]\Big).$$

The second term is non-negative by (B). For the first, fix $A = u$. If $u \leq 0$, then $uB \leq 0 \leq c$, so $\pi = 0$ and the conditional covariance is zero. If $u > 0$, then $\pi = \mathbf{1}\{B > c/u\}$ is a non-decreasing function of $B$, and since $\pi$ is a function of $(A, B)$, $\mathrm{Cov}(\pi, \bar{H} \mid A = u) = \mathrm{Cov}(\pi, \psi(u, B) \mid A = u)$. Two non-decreasing functions of the same random variable have non-negative covariance (Chebyshev's association inequality), so the first term is non-negative, and (i) gives $\eta(\pi) \geq \mathbb{E}[H_i]$. Under the strictness condition, the conditional covariance is strictly positive on a set of $u$ of positive probability. □

*Proof of Corollary 2*

*Proof.* For any $\pi$ satisfying both constraints,

$$\mathcal{W}(\pi^\ddagger) - \mathcal{W}(\pi) = \mathbb{E}\Big[(\pi^\ddagger - \pi)(s - \mu^* \hat{h} - \lambda^*)\Big] + \lambda^*\Big(\mathbb{E}[\pi^\ddagger] - \mathbb{E}[\pi]\Big) + \mu^*\Big(\mathbb{E}[\pi^\ddagger \hat{h}] - \mathbb{E}[\pi \hat{h}]\Big).$$

The first integrand is non-negative pointwise, as in Proposition 1. By complementary slackness the second term equals $\lambda^*(\kappa - \mathbb{E}[\pi]) \geq 0$ and the third equals $\mu^*(\eta_{\max}\kappa - \mathbb{E}[\pi \hat{h}]) \geq 0$. If $\pi$ is also optimal, the first term vanishes, so $\pi = \pi^\ddagger$ almost surely off the boundary set $\{s - \mu^* \hat{h} = \lambda^*\}$, which has probability zero. □

## Appendix B. Risk-Modulation Function, Example and Frontier

This appendix gives the piecewise linear preference weight $g$ used in the simulation, a numerical example of the three scoring rules, and two remarks on how $g$ trades operator welfare against the targeting pattern. The functional form is an implementation choice; the parameters are chosen for their operational reading.

*The preference weight*

For retention bonuses, the effectiveness of a bonus depends on the player's churn risk $p$ (we write $p$ for $R$ here). Three regimes motivate the shape. Below $p_L$ the player is stable and would likely stay without a bonus, so the weight is zero. Between the plateau bounds the player is at risk but reachable, so the weight is one. Above $p_H$ the player has probably disengaged beyond what a bonus can reverse (Ascarza, 2018), so the weight falls to a floor $g_{\min}$ that reflects uncertainty in the churn estimate. The parameters $\theta = (p_L, p_{ML}, p_{MH}, p_H, g_{\min})$ have these direct meanings, and Figure B.3 plots $g$ at the default $\theta_0 = (0.05, 0.40, 0.60, 0.90, 0.20)$ used in the simulation. The defaults are chosen for clear contrasts across regimes, and an operator should set $\theta$ from domain evidence. The weight $g$ is a covariate-dependent targeting weight on an observable churn estimate, not risk aversion in the decision-theoretic sense of Manski and Tetenov (2007).

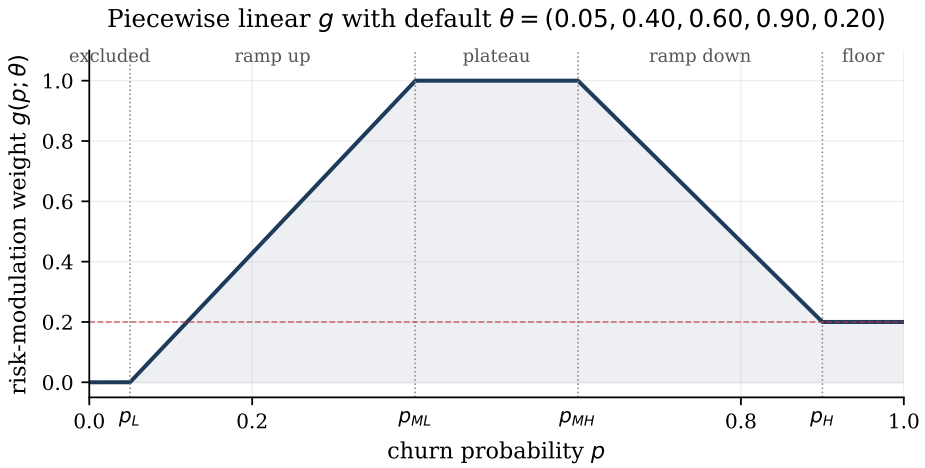



Figure B.3: The piecewise linear preference weight at default parameters $\theta_0 = (0.05, 0.40, 0.60, 0.90, 0.20)$. Players below $p_L$ are excluded. The weight ramps to one through the moderate-risk plateau and falls to the floor $g_{\min}$ for the highest-churn players.

**Definition 2** (Piecewise linear risk-modulation function)**.** *The risk-modulation function* $g : [0,1] \to [0,1]$ *is defined as*

$$g(p;\,\theta) = \begin{cases} 0 & \text{if } p < p_L, \\ \dfrac{p - p_L}{p_{ML} - p_L} & \text{if } p_L \le p < p_{ML}, \\ 1 & \text{if } p_{ML} \le p \le p_{MH}, \\ 1 - (1 - g_{\min})\,\dfrac{p - p_{MH}}{p_H - p_{MH}} & \text{if } p_{MH} < p < p_H, \\ g_{\min} & \text{if } p \ge p_H, \end{cases} \tag{B.1}$$

*parameterised by* $\theta = (p_L, p_{ML}, p_{MH}, p_H, g_{\min})$ *with* $0 \le p_L < p_{ML} \le p_{MH} < p_H \le 1$ *and* $g_{\min} \in [0,1]$.

**Remark 8** (Properties of the piecewise linear $g$)**.** *The function* $g$ *defined in* (B.1) *satisfies: (i)* $g$ *is continuous on* $[0,1]$*; (ii)* $g$ *is non-decreasing on* $[0, p_{MH}]$ *and non-increasing on* $[p_{MH}, 1]$*; (iii)* $g(p) = 0$ *for* $p < p_L$ *and* $g(p) \ge g_{\min}$ *for all* $p \ge p_H$*; on the ascending ramp* $[p_L, p_{ML})$*,* $g(p)$ *can be below* $g_{\min}$ *when* $g_{\min} > 0$*; (iv)* $g$ *is Lipschitz continuous, so expectations involving it are well-behaved. The five-parameter form is an implementation choice, not part of the contribution; any bounded, measurable weight would serve, and the parameters are chosen for their direct operational reading (above) rather than for approximation-theoretic reasons.*

**Remark 9** (Welfare-risk frontier)**.** *Each choice of* $g$ *gives a pair of operator welfare and risk exposure. Observation 1 places the neutral weight* $g \equiv 1$ *at the highest-welfare point of the resulting curve, and it does not make the curve monotone in any parameter of* $g$*.*

**Remark 10** (When $g$ could offset an estimation error)**.** *Observation 1 concerns oracle-optimal rules. If the true effect factors as* $\tau(x) = \tau_0(x)\, m(R(x))$ *and the effect model cannot capture the moderation* $m$*, a weight* $g = m$ *could improve an estimated rule. This requires an operator-supplied* $g$ *that mirrors a structural interaction no estimated component recovers, and the simulated misspecification (dropped features in the effect model) is*

*not of that kind. The regime is a theoretical possibility, not a claim about the simulated data.*

*Numerical example*

**Example 1** (Illustrative targeting)**.** *Eight hypothetical players have known treatment effects, operator values and churn risks (Table B.9). The values are abstract scores chosen so that three scoring rules select visibly different sets, and they do not respect the bound $\tau \leq p$ that a strict binary-retention reading would impose. We use $s_i = \tau_i V_i \, g(p_i; \theta_0)$ with $\theta_0 = (0.05, 0.40, 0.60, 0.90, 0.20)$ and a budget of 3 of 8 players.*

Table B.9: Numerical example with eight hypothetical players: treatment effect $\tau$ (a probability), operator value $V$, churn risk $p$, preference weight $g(p)$, score $s$, and the targeting rank under three scoring rules. Budget: 3 players.

| Player | $\tau$ | $V$ | $p$ | $g(p)$ | $s$ | Rank: $\tau$ | Rank: $\tau V$ | Rank: $\tau V g$ |
|---|---|---|---|---|---|---|---|---|
| A | 0.35 | 200 | 0.03 | 0.00 | 0.0 | 1 | 4 | 7 |
| B | 0.30 | 500 | 0.25 | 0.57 | 85.7 | 2 | 2 | 3 |
| C | 0.20 | 800 | 0.50 | 1.00 | 160.0 | 3 | 1 | 1 |
| D | 0.15 | 600 | 0.45 | 1.00 | 90.0 | 4 | 3 | 2 |
| E | 0.10 | 400 | 0.80 | 0.47 | 18.7 | 5 | 6 | 4 |
| F | 0.05 | 950 | 0.10 | 0.14 | 6.8 | 7 | 5 | 5 |
| G | −0.10 | 900 | 0.50 | 1.00 | −90.0 | 8 | 8 | 8 |
| H | 0.08 | 300 | 0.95 | 0.20 | 4.8 | 6 | 7 | 6 |

*Ranking on $\tau$ alone selects A, B and C. Player A has the largest effect but a churn risk of 0.03, so a bonus is wasted on a player who would stay anyway. Ranking on $\tau V$ selects C, B and D, because A's low value of 200 drops A to fourth. The full score selects C, D and B: A is vetoed by $g = 0$, E is down-weighted for high churn, and C and D sit on the plateau. Player G is made worse off by treatment, and all three rules exclude G. None of the three rules considers harm, which is the subject of Section 5.*

*Welfare-risk frontier in the simulation*

Table B.10 and Figure B.4 report operator welfare and risk exposure across fixed configurations of $g$, the calibrated configuration and an inverted $g$, at $\kappa = 0.30$ with welfare scored under $g \equiv 1$. Every fixed non-identity configuration sits below the neutral weight in welfare, consistent with, though not implied by, Observation 1, which covers oracle rules. The inverted $g$, which puts high weight at extreme churn and low weight in the moderate zone, earns 13,934 at risk exposure 0.145 and is also below the neutral weight. The calibrated row is the AIPW-calibrated rule, which does not reproduce the identity and sits below it (Appendix Appendix C). Risk exposure varies far less across configurations than welfare does, so in this simulation the dial on $g$ is mostly a welfare cost and only weakly reshapes who is treated. A population in which churn risk clusters more sharply in the treated tail would show wider variation in exposure.

## Appendix C. Calibration of the Preference Weight

This appendix describes how the preference weight is calibrated in the simulation and what the calibration shows. Each replication splits its training set 60/40 into an

Table B.10: Welfare and risk exposure under different configurations of $g$ ($\kappa = 0.30$, welfare scored under $g \equiv 1$). Labels show $(p_L/p_{ML}/p_{MH})$, with $p_H = 0.90$ and $g_{\min} = 0.20$ unless stated. Calibrated is the AIPW-calibrated configuration. Risk exposure is mean churn probability among the treated.

| Configuration | Welfare | Risk exposure |
|---|---|---|
| No reg ($g$=1) | 16,446 | 0.136 |
| Wide (0.03/0.50/0.70) | 12,599 | 0.148 |
| Default (0.05/0.40/0.60) | 13,646 | 0.146 |
| Narrow (0.10/0.30/0.50) | 14,935 | 0.145 |
| Aggressive (0.15/0.35/0.50/$g_{\min}$=0.05) | 13,738 | 0.149 |
| Calibrated | 15,566 | 0.142 |
| Inverted $g$ | 13,934 | 0.145 |

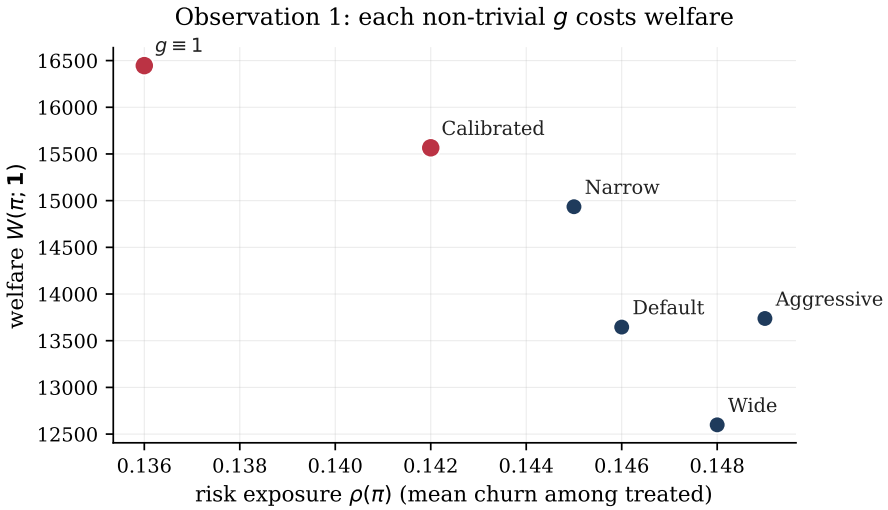



Figure B.4: Welfare-risk frontier: each fixed configuration of $g$ sits below the neutral weight $g \equiv 1$ in welfare. This is consistent with, though not implied by, Observation 1, which covers oracle rules.

estimation fold, which fits the component models, and a validation fold, which scores candidate configurations of $\theta$. The candidate set has 22 configurations, including a near-identity configuration ($p_L = 0$, $p_{ML} = 0.01$, $p_{MH} = 0.99$, $g_{\min} = 1$) that makes $g$ constant on $[0.01, 0.99]$. Because $\hat{p} \geq 0.01$, this configuration is equivalent to the identity, and the two select the same set. Each candidate rule treats the top $\kappa$ share of the validation fold by $\hat{\tau}\hat{V}\, g(\hat{p}; \theta)$, and the objective scores the selected set. Ties go to the lowest grid index, and no oracle information enters the selection.

The plug-in objective sums $\hat{\tau}\hat{V}$ over the selected set. The identity candidate selects exactly the set that maximises this sum, so the plug-in objective picks the identity, or a candidate with the same plug-in value, by construction (Observation 1). It selects the identity in 200 of 200 replications with default learners and in 198 with tuned learners, and its welfare difference from the identity is exactly zero, so it has no $p$-value. These rows are degenerate and carry no evidence about whether $g$ helps.

The AIPW objective sums $\hat{\Gamma}_i \hat{V}_i$ over the selected set, where $\hat{\Gamma}_i$ is the cross-fitted doubly robust score of the effect under the known propensity 0.5. Nothing forces it to pick the identity, and it selects the identity in 13 of 200 replications (6.5%) with default learners and 13 (6.5%) with tuned learners. Table C.11 shows that AIPW calibration costs operator welfare against the identity: $-881$ [$-1{,}217$, $-544$], $d = -0.37$, or $-5.4\%$ [$-7.4\%$, $-3.3\%$], with default learners, and $-373$ [$-667$, $-78$], $d = -0.18$, with tuned learners. Regret against the feasible oracle is 17,231 for AIPW calibration and 16,350 for the plug-in rule with default learners. One reading, which we did not test, is that the validation fold is small and the AIPW objective noisy, so the maximum over 22 candidates tends to pick a configuration that looks good on that fold by chance. No configuration in the grid beats the identity on true welfare in this DGP, and the fixed configurations of Appendix Appendix B are all below it as well.

Table C.11: Calibration of the preference weight ($\kappa = 0.30$, 200 replications). Identity rate and count are the share and number of replications in which calibration selected the identity configuration. Difference is calibrated minus identity welfare with a 95% confidence interval, and regret is against the feasible oracle with a 95% confidence interval. † The plug-in objective selects the identity by construction, so its difference is exactly zero and it has no $p$-value.

| Objective | Learners | Identity rate | Identity count | Calibrated minus identity | Holm $p$ | Regret vs feasible oracle |
|---|---|---|---|---|---|---|
| AIPW | default | 6.5% | 13/200 | −881 [−1,217, −544] | <0.001 | 17,231 [16,669, 17,792] |
| Plug-in (degenerate†) | default | 100.0% | 200/200 | +0 [0, 0] | – | 16,350 [15,917, 16,784] |
| AIPW | tuned | 6.5% | 13/200 | −373 [−667, −78] | 0.0272 | 16,228 [15,347, 17,109] |
| Plug-in | tuned | 99.0% | 198/200 | +4 [−3, 10] | 0.274 | 15,852 [14,998, 16,706] |

## Appendix D. Harm Sweep Welfare

Table D.12 gives operator welfare for every cell of the harm sweep of Section 6.2. The harm-blind rule's welfare does not depend on $c_{\text{harm}}$ or $\sigma_h$ because it never reads harm and welfare counts only $\tau_i V_i$. Welfare differences between the three estimated rules are largest at $c_{\text{harm}} = 1$. The population mean harm falls from 0.458 at $c_{\text{harm}} = -1$ to 0.421 at $c_{\text{harm}} = 1$, and the share of players above the soft weight's lower threshold 0.6 falls from 29.0% to 20.8%. The mean drifts although the variance of the harm logit is fixed, because $z_{\text{spend}}$ is right-skewed. Because the share of players the soft weight can down-weight differs across rows, welfare differences across $c_{\text{harm}}$ mix the effect of alignment with the effect of

this shift in the harm distribution. The soft weight also beats harm-blind ranking at $c_{\text{harm}} = -1$ at every $\sigma_h$ (Table D.12), which fits the reading that $\hat{H}$ proxies structural value: where harm falls with value, a high $\hat{H}$ flags low value.

Table D.12: Operator welfare in the harm sweep ($\kappa = 0.30$). Columns follow Table 2.

| | | Estimated components | | | Feasible-oracle components | | |
|---|---|---|---|---|---|---|---|
| $c_{\text{harm}}$ | $\sigma_h$ | Harm-blind | Soft weight | Hard cap | Harm-blind | Soft weight | Hard cap |
| −1.0 | 0.0 | 16,446 | 16,915 | 16,617 | 32,797 | 32,496 | 32,707 |
| −1.0 | 0.5 | 16,446 | 16,839 | 16,600 | 32,797 | 32,215 | 32,533 |
| −1.0 | 1.0 | 16,446 | 16,615 | 16,487 | 32,797 | 31,459 | 32,086 |
| −0.5 | 0.0 | 16,446 | 16,381 | 16,355 | 32,797 | 31,455 | 32,079 |
| −0.5 | 0.5 | 16,446 | 16,363 | 16,318 | 32,797 | 31,275 | 31,930 |
| −0.5 | 1.0 | 16,446 | 16,297 | 16,193 | 32,797 | 30,772 | 31,566 |
| 0.0 | 0.0 | 16,446 | 16,387 | 16,189 | 32,797 | 31,094 | 31,851 |
| 0.0 | 0.5 | 16,446 | 16,377 | 16,156 | 32,797 | 30,895 | 31,704 |
| 0.0 | 1.0 | 16,446 | 16,420 | 16,033 | 32,797 | 30,483 | 31,359 |
| 0.5 | 0.0 | 16,446 | 17,537 | 16,058 | 32,797 | 31,156 | 31,901 |
| 0.5 | 0.5 | 16,446 | 17,516 | 16,015 | 32,797 | 30,961 | 31,771 |
| 0.5 | 1.0 | 16,446 | 17,397 | 15,893 | 32,797 | 30,454 | 31,380 |
| 1.0 | 0.0 | 16,446 | 19,045 | 15,701 | 32,797 | 31,986 | 32,361 |
| 1.0 | 0.5 | 16,446 | 18,897 | 15,703 | 32,797 | 31,663 | 32,185 |
| 1.0 | 1.0 | 16,446 | 18,582 | 15,669 | 32,797 | 30,909 | 31,719 |

## Appendix E. Additional Scenarios and Tuning Details

*Scenarios*

Table E.13 reports the comparison of CATE × LTV with CATE only, and a few other contrasts, in each additional scenario at $\kappa = 0.30$ unless stated. The first difference column uses the default T-learner and the last two columns repeat the comparison with the tuned T-learner, which was tuned on the main design only and reused unchanged. The scenarios are the main design; a null design with no effect-churn interaction ($m(p) \equiv 1$); correlated covariates drawn through a Gaussian copula; a misspecified effect model that drops login frequency while the value and churn models keep all features; an asymmetric moderation $m(p) \propto p(1-p)^3$ that peaks near $p = 0.25$; a correlation sweep with $V \leftarrow V + \gamma \cdot 180\, z(\tau_{\text{base}})$; a heavy-tailed value $V$ with a multiplicative log-normal term, clipped at 100,000; and an adversarial design in which the effect also depends on bonus history, which does not enter $V$, and the effect model drops it.

*Misspecified effect model and the adversarial case..* With login frequency dropped from the effect model, CATE × LTV beats CATE only by +287 [170, 404], $d = 0.34$, with the default T-learner, because login frequency also enters value and the value model therefore carries part of the lost signal. With the tuned T-learner the sign reverses, to −2,265 [−2,666, −1,865], $d = -0.79$, so the rescue depends on the learner. In the adversarial design the value model cannot recover the dropped signal, and value weighting does not help: at $\kappa = 0.30$ the difference is −133 [−302, 36], and at $\kappa = 0.10$ it is −1,262 [−1,426,

Table E.13: Additional scenarios (200 replications, $\kappa = 0.30$ unless named). Difference is the first named strategy minus the second, with a 95% confidence interval and Cohen's $d$. For like-for-like rows it is all features minus the dropped feature, so a positive value is a loss from dropping. Family is the size of the scenario's Holm family. The last two columns repeat the comparison with the tuned T-learner where one exists. The nominal-versus-effective row is nominal minus clip-adjusted welfare.

| Scenario | Comparison | Difference | 95% CI | $d$ | Holm $p$ | Family | Tuned difference [95% CI] | Tuned Holm $p$ |
|---|---|---|---|---|---|---|---|---|
| Main DGP | CATE × LTV vs CATE Only | −187 | [−319, −55] | −0.20 | 0.0170 | 24 | −1,819 [−2,312, −1,326] | <0.001 |
| Null DGP | CATE × LTV vs CATE Only | +5,157 | [4,704, 5,610] | 1.59 | <0.001 | 24 | +191 [−1,988, 2,370] | 0.863 |
| Correlated DGP | CATE × LTV vs CATE Only | −678 | [−800, −556] | −0.77 | <0.001 | 24 | −4,215 [−4,795, −3,636] | <0.001 |
| Misspecified CATE model | CATE × LTV vs CATE Only | +287 | [170, 404] | 0.34 | <0.001 | 24 | −2,265 [−2,666, −1,865] | <0.001 |
| Asymmetric DGP | CATE × LTV vs CATE Only | +398 | [152, 644] | 0.23 | 0.0016 | 8 | −2,716 [−3,750, −1,682] | <0.001 |
| Corr. sweep, $\gamma = -1$ | CATE × LTV vs CATE Only | −123 | [−258, 12] | −0.13 | 0.0742 | 18 | −946 [−1,308, −584] | <0.001 |
| Corr. sweep, $\gamma$=0 | CATE × LTV vs CATE Only | −329 | [−464, −194] | −0.34 | <0.001 | 18 | −1,664 [−2,098, −1,231] | <0.001 |
| Corr. sweep, $\gamma$=1 | CATE × LTV vs CATE Only | −504 | [−647, −362] | −0.49 | <0.001 | 18 | −2,390 [−2,930, −1,850] | <0.001 |
| Heavy-tailed DGP | CATE × LTV vs CATE Only | −733 | [−1,041, −425] | −0.33 | <0.001 | 6 | −2,910 [−3,698, −2,123] | <0.001 |
| Adversarial dropout, $\kappa$=0.30 | CATE × LTV vs CATE Only | −133 | [−302, 36] | −0.11 | 0.319 | 16 | −1,739 [−2,270, −1,208] | <0.001 |
| Adversarial dropout, $\kappa$=0.10 | CATE × LTV vs CATE Only | −1,262 | [−1,426, −1,098] | −1.07 | <0.001 | 16 | −2,994 [−3,320, −2,669] | <0.001 |
| Nominal vs effective welfare | CATE × LTV vs CATE × LTV [effective] | +787 | [743, 832] | 2.47 | <0.001 | 17 | – | – |
| Well-specified vs misspecified V | CATE × LTV, well-specified V vs CATE × LTV, misspecified V | −532 | [−573, −491] | −1.81 | <0.001 | 2 | – | – |
| AIPW calibration vs identity | Calibrated (aipw) vs Identity (g=1) | −881 | [−1,217, −544] | −0.37 | <0.001 | 3 | – | – |
| Like-for-like, login dropped | CATE Only (all features) vs CATE Only (feature dropped) | +9,934 | [9,554, 10,314] | 3.65 | <0.001 | 5 | – | – |
| Like-for-like, bonus dropped | CATE Only (all features) vs CATE Only (feature dropped) | +13,636 | [13,207, 14,066] | 4.43 | <0.001 | 5 | – | – |
| CATE share, misspec vs wellspec | CATE share, misspec vs well-spec [T, plug-in] | +1.5 pp | [1.4, 1.7] pp | 1.71 | <0.001 | 6 | – | – |

−1,098], $d = -1.07$, or −24.3% [−27.7%, −21.0%]. The tuned T-learner never splits on bonus history, so dropping it changes nothing for that learner. The tuned adversarial rows therefore compare value weighting under a learner that ignores the adversarial feature, not the effect of dropping it.

*Null design..* With no effect-churn interaction, CATE × LTV beats CATE only by +5,157 [4,704, 5,610], $d = 1.59$, with the default T-learner. With the tuned T-learner the difference is +191 [−1,988, 2,370], whose interval includes zero, so the null-design advantage of value weighting holds for the default T-learner only.

*Feature dropout like for like..* Dropping login frequency from the effect model costs CATE only +9,934 [9,554, 10,314], a loss of +59.7%, and costs the S-learner +15,441 [14,704, 16,178], a loss of +63.6%. Dropping bonus history in the adversarial design costs CATE only +13,636 [13,207, 14,066]. The S-learner is not more robust to losing an informative feature; its advantage is that it uses a given feature set more effectively, and comparing it with a rule whose effect model lost the feature is not like for like, which is why learner choice is a separate question. An observational diagnostic bounds the mean absolute score gap over the misranked set and splits the bound into an effect term and a value term in one fixed telescoping order. It is not a regret share, because only the Shapley split of Remark 5 decomposes regret. The per-replication mean effect share is 93.1% [92.9%, 93.2%] for the full-feature T-learner against a held-out plug-in reference, and the share rises by +1.5*pp* [1.4*pp*, 1.7*pp*] when login is dropped. Against the feasible and hindsight oracles the same full-feature share is 76.9% [76.7%, 77.1%] and 79.5% [79.3%, 79.6%], a score-error heuristic that is lower than the Shapley share of Section 6.6.

*Other scenarios..* Under the asymmetric moderation CATE × LTV beats CATE only by +398 [152, 644], whereas on the main design the sign is negative (−187), and under the tuned T-learner the asymmetric gain reverses to −2,716 [−3,750, −1,682], so whether value weighting helps at a given budget depends on a moderation shape the operator cannot observe in advance. In the correlation sweep the difference is −123 [−258, 12],

$-329$ [$-464$, $-194$] and $-504$ [$-647$, $-362$] for $\gamma = -1$, 0 and 1, with the same sign and a larger magnitude under the tuned T-learner. With a heavy-tailed value the difference is $-733$ [$-1{,}041$, $-425$], or $-2.7\%$ of CATE only. Nominal minus clip-adjusted CATE $\times$ LTV welfare is $+787$ [743, 832], $d = 2.47$, so reported welfare is nominal and overstates realised lift. Replacing the proxy value model with the well-specified one changes CATE $\times$ LTV by $-532$ [$-573$, $-491$], the proxy-beats-well-specified result of Section 6.6.

*Learner tuning*

Each learner is tuned once, on the main design, by mean cross-fitted R-loss over 5 pilot replications, and the tuned parameters are then fixed for every study. The grid has 80 configurations: depth 1 to 4, 25, 50, 100 or 300 trees, and learning rate 0.005, 0.01, 0.02, 0.05 or 0.1. Within a pilot replication the training split is cross-fitted into two folds, the learner is fitted on one and scored on the other by mean$\Big((Y - \hat{m} - (W - 0.5)\,\hat{\tau})^2\Big)$ with $\hat{m}$ a fixed gradient-boosted estimate of $\mathbb{E}[Y \mid X]$. The pilot seeds are disjoint from the evaluation seeds. The chosen configurations are depth 1, 25 trees and rate 0.020 for the T-learner; depth 2, 100 trees and rate 0.010 for the X-learner; depth 2, 300 trees and rate 0.100 for the S-learner; and depth 1, 100 trees and rate 0.005 for the DR-learner. The T-learner chooses the smallest depth and tree count, the DR-learner the smallest depth and learning rate, and the S-learner the largest tree count and learning rate, so all three tuned fits may be limited by the grid. For T and DR the edge is toward more shrinkage, which strengthens the attenuation reading, although a wider grid could change the magnitudes. The chosen R-loss is 0.21258 (T), 0.21255 (X), 0.21253 (S) and 0.21257 (DR), a spread across learners of 0.00005, while the worst-to-best range across the grid is 0.01573 (T), 0.00803 (X), 0.00021 (S) and 0.02108 (DR). The S-learner's loss is almost flat across its grid, so its tuned configuration is weakly identified. R-loss measures how well the estimate calibrates against residualised outcomes and does not measure how well it ranks players, so a lower R-loss need not mean better targeting.

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